LOGIC.SEARCH.01 · Searches with many cases

Every whole number of cents from 8 to 30 can be paid exactly using only 3-cent and 5-cent stamps.

Answer: True

Reasoning: 8 = 3 + 5, 9 = 3 + 3 + 3, 10 = 5 + 5. Once three amounts in a row work, adding a 3-cent stamp reaches every larger amount.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S36-F2-Q5 · question page 846

LOGIC.SEARCH.01 · Searches with many cases

Give a whole number of cents from 1 to 10 that CANNOT be paid exactly using only 3-cent and 5-cent stamps.

Answer: any values that satisfy: n == 1 or n == 2 or n == 4 or n == 7 (example: n = 7)

Reasoning: Checking 1 to 10: 3, 5, 6, 8, 9 and 10 can be made; 1, 2, 4 and 7 cannot.

Tempting answers the rule rejects: n = 8; n = 6; n = 10; n = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 4 listed mistakes fail it.

S37-M-Q1 · question page 848

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the sum of two odd integers is even in order.

Answer: Let m = 2a + 1 and n = 2b + 1, where a and b are integers. → Then m + n = 2a + 2b + 2. → So m + n = 2(a + b + 1). → Since a + b + 1 is an integer, m + n is even.

Reasoning: Start from the definitions, add, factor out 2, then name the integer that shows the sum is even.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S37-M-Q2 · question page 849

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which first line correctly begins a direct proof that the product of any two odd integers is odd?

Answer: C. Let m = 2a + 1 and n = 2b + 1, where a and b are integers.

Reasoning: Two general odd integers need two separate integers a and b.

- ✗ Let m = 2a + 1 and n = 2a + 1, where a is an integer.: using the same a makes m and n equal, so it proves the claim only for a number times itself
- ✗ Let m = 3 and n = 5.: one example cannot prove a claim about all odd integers
- ✗ Assume that m times n is odd.: that assumes the conclusion the proof must reach

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-M-Q3 · question page 850

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A proof that an even integer plus an odd integer is odd reaches the line m + n = 2a + 2b + 1. Which line correctly finishes it?

Answer: D. m + n = 2(a + b) + 1, and a + b is an integer, so m + n is odd.

Reasoning: The definition of odd needs the form 2(integer) + 1.

✗ m + n is odd because the expression 2a + 2b + 1 contains a 1.: containing a 1 is not the definition of odd

✗ Since 4 + 3 = 7 is odd, m + n is odd.: an example does not finish a general proof

✗ m + n = 2(a + b + 1), so m + n is even.: 2a + 2b + 1 is not equal to 2(a + b + 1)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-M-Q4 · question page 851

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Checking that 3 + 5, 7 + 9 and 11 + 13 are all even proves that the sum of any two odd integers is even.

Answer: False

Reasoning: Examples can support a guess but cannot prove a claim about infinitely many integers. A proof must use general odd integers 2a + 1 and 2b + 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-M-Q5 · question page 852

LOGIC.PROOF.01 · Build a direct argument about even, odd and multiples from definitions

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. To show that 6n + 4 is even for every integer n, write 6n + 4 = 2(pn + q). Give the numbers p and q.

Answer: any values that satisfy: p == 3 and q == 2 (example: p = 3, q = 2)

Reasoning: 6n + 4 = 2(3n + 2), and 3n + 2 is an integer, so 6n + 4 is even.

Tempting answers the rule rejects: p = 6, q = 4; p = 3, q = 4; p = 2, q = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S37-B1-Q1 · question page 853

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which expression is odd for every integer k?

Answer: A. 2k + 7

Reasoning: 2k + 7 = 2(k + 3) + 1, which has the odd form.

- ✗ 2k + 4: 2k + 4 = 2(k + 2), which is even
- ✗ 4k: 4k = 2(2k), which is even
- ✗ k + 1: k + 1 is odd only when k is even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-B1-Q2 · question page 854

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some

integer k. An integer n is a multiple of m if n = mk for some integer k. 2k + 6 is even for every integer k.

Answer: True

Reasoning: 2k + 6 = 2(k + 3).

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-B1-Q3 · question page 855

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Write 19 in the form 2k + 1. What is k?

Answer: 9

Reasoning: 19 = 2(9) + 1.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S37-B1-Q4 · question page 856

LOGIC.PROOF.01 · Use the definitions of even and odd

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n = 2k + 1, then n + 1 is odd.

Answer: False

Reasoning: n + 1 = 2k + 2 = 2(k + 1), which is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S37-B1-Q5 · question page 857

LOGIC.PROOF.01 · Use the definitions of even and odd

Which statement is the definition of an even integer?

Answer: D. n = 2k for some integer k

Reasoning: Even means 2 times some integer.

- ✗ n ends in the digit 0: that describes multiples of 10; 4 is even but ends in 4
- ✗ n = 2k + 1 for some integer k: that is the definition of odd
- ✗ n leaves remainder 1 when divided by 2: that also describes odd numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F1-Q1 · question page 858

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that if n is a multiple of 3, then n squared is a multiple of 9 in a correct order. (Two orders are correct.)

Answer: Let n = 3k, where k is an integer. → Then n squared = 9k squared. → k squared is an integer, because k is an integer. → So n squared is 9 times an integer, a multiple of 9.

Also accepted: Let n = 3k, where k is an integer. → k squared is an integer, because k is an integer. → Then n squared = 9k squared. → So n squared is 9 times an integer, a multiple of 9.

Reasoning: Begin with the definition (A) and end with the conclusion (D). B and C can come in either order.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S37-F1-Q2 · question page 859

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Which is a correct direct proof that the sum of three consecutive integers is a multiple of 3?

Answer: A. n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1), and n + 1 is an integer.

Reasoning: Write three general consecutive integers, add, and factor out 3.

- ✗ 1 + 2 + 3 = 6, and 6 is a multiple of 3.: one example does not prove the general claim
- ✗ n + n + n = 3n, which is a multiple of 3.: n, n, n are not consecutive integers
- ✗ n + (n + 1) + (n + 2) = 3n + 3, which is odd.: 3n + 3 is not always odd, and odd is not the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F1-Q3 · question page 860

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. If n is odd, then n squared minus 1 is a multiple of 8.

Answer: True

Reasoning: n = 2k + 1 gives n squared - 1 = 4k(k + 1). One of k and k + 1 is even, so k(k + 1) = 2j and the expression is 8j.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S37-F1-Q4 · question page 861

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. (2a + 1)(2b + 1) = 2(2ab + a + b) + 1. For a = 3 and b = 4, what number is 2ab + a + b?

Answer: any values that satisfy: x == 31 (example: x = 31)

Reasoning: 2(3)(4) + 3 + 4 = 31. Check: 7 x 9 = 63 = 2(31) + 1.

Tempting answers the rule rejects: x = 63; x = 24; x = 32

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S37-F1-Q5 · question page 862

LOGIC.PROOF.01 · Direct proofs about multiples

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. A student proves that every multiple of 4 is even by writing n = 4k = 2(2k). What justifies the last sentence, "so n is even"?

Answer: B. 2k is an integer, so n is 2 times an integer.

Reasoning: The definition needs n = 2 times an integer, and 2k is an integer.

- ✗ 4 is an even number.: that alone does not put n in the form 2 times an integer
- ✗ n can be any number at all.: the claim is about multiples of 4 only
- ✗ k must be an even number.: k can be any integer; nothing makes it

even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F2-Q1 · question page 863

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Put the steps of a direct proof that the square of an odd integer leaves remainder 1 when divided by 4 in order.

Answer: Let n = 2k + 1, where k is an integer. → Then n squared = 4k squared + 4k + 1. → So n squared = 4(k squared + k) + 1. → Since k squared + k is an integer, n squared leaves remainder 1 when divided by 4.

Reasoning: Definition, expand, factor out 4, then name the integer.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S37-F2-Q2 · question page 864

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Claim: the sum of two even integers is a multiple of 4. "Proof: Let m = 2k and n = 2k. Then m + n = 4k." What is wrong?

Answer: C. It uses the same k for both, so it only covers the case m = n.

Reasoning: With separate integers, m + n = 2(a + b), which is a multiple of 4 only when a + b is even. 2 + 4 = 6 shows the claim is false.

✗ It should have started with m = 4k.: the given numbers are only known to be even

✗ Nothing is wrong; the proof is correct.: 2 + 4 = 6 is not a multiple of 4, so the claim is false

✗ It should check an example before starting.: checking examples is not the flaw

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S37-F2-Q3 · question page 865

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. The statement "n squared + n is even for every integer n" can be proved by two cases: n even and n odd.

Answer: True

Reasoning: If n = 2k, n squared + n = 2(2k squared + k). If n = 2k + 1, n squared + n = (2k + 1)(2k + 2) = 2(2k + 1)(k + 1). Both are even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S37-F2-Q4 · question page 866

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. For odd n = 2k + 1, n squared + 4n + 3 = 2(2k squared + 6k + 4). For k = 5, what number is 2k squared + 6k + 4?

Answer: 84

Reasoning: 2(25) + 30 + 4 = 84. Check: n = 11 gives 121 + 44 + 3 = 168 = 2 x 84.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S37-F2-Q5 · question page 867

LOGIC.PROOF.01 · Judge and complete direct proofs

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. Give an integer n from 1 to 20 for which n squared + n + 1 is even. If no such n exists, enter 0.

Answer: any values that satisfy: n == 0 (example: n = 0)

Reasoning: n squared + n = n(n + 1) is always even, so n squared + n + 1 is always odd. No such n exists.

Tempting answers the rule rejects: n = 1; n = 2; n = 7

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S38-M-Q1 · question page 869

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many different ways can a rectangle 2 squares tall and 4 squares wide be tiled with dominoes?

Answer: 5

Reasoning: Look at the left end: one upright domino leaves a 2 by 3 (3 ways), or two flat dominoes leave a 2 by 2 (2 ways): 3 + 2 = 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-M-Q2 · question page 870

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. The 3 by 3 board with its center square removed (shown) can be tiled by 4 dominoes.

Answer: True

Reasoning: The 8 remaining squares form a ring; going around it, pair the squares two at a time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-M-Q3 · question page 871

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. Which region can be tiled by 4 dominoes?

Answer: B. Figure B

Reasoning: Region B: the top-left square must pair with the square below it, and the rest then pairs off in one way.

- ✗ Figure A: color it like a checkerboard: 5 squares of one color and 3 of the other, but every domino covers one of each
- ✗ Figure C: color it like a checkerboard: 5 squares of one color and 3 of the other
- ✗ Figure D: the right end of the long row and the square below the row each touch only one square, the same one, so both cannot be covered

Grading: The chosen letter must equal the keyed letter.

Verified: Automatic: the panel check accepts only the keyed panel.

S38-M-Q4 · question page 872

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 5 by 5 board, and the other 24 squares are to be tiled by 12 dominoes. Rows and columns are numbered 1 to 5. Give the row r and column c of a square whose removal makes a tiling possible.

Answer: any values that satisfy: isInteger(r) and isInteger(c) and r >= 1 and r <= 5 and c >= 1 and c <= 5 and isEven(r + c) (example: r = 1, c = 1)

Reasoning: Color the board like a checkerboard with the corners dark: 13 dark, 12 light. A tiling needs 12 of each, so the removed square must be dark (r + c even). Every dark square works; a computer search of all 25 removals confirms it.

Tempting answers the rule rejects: r = 1, c = 2; r = 2, c = 3; r = 6, c = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Tilings counted for every one of the 25 removals: exactly the 13 squares with r + c even leave a tileable board. Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 13 valid answers.

S38-M-Q5 · question page 873

LOGIC.SPATIAL.05 · Decide which pieces cover a region exactly, and prove when no covering exists

An L-tromino is three squares in an L shape (a 2 by 2 square with one square missing). It may be turned any way. In how many ways can a rectangle 2 squares tall and 3 wide be tiled by two L-trominoes?

Answer: 2

Reasoning: One L must cover the top-left square. Of its placements, only two leave an L-shaped gap; each gives one tiling.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q1 · question page 874

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A 4 by 4 board is colored like a checkerboard. How many of its squares are dark?

Answer: 8

Reasoning: Each row has 2 dark and 2 light squares: 4 x 2 = 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q2 · question page 875

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

Every domino placed on a checkerboard covers one dark square and one light square.

Answer: True

Reasoning: The two squares share a side, and squares sharing a side have different colors.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S38-B1-Q3 · question page 876

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 7 dark and 5 light squares on a checkerboard coloring. According to the color count, what is the greatest number of dominoes that could be placed on it without overlapping?

Answer: 5

Reasoning: Each domino uses one light square, and there are only 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-B1-Q4 · question page 877

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region has 6 dark and 6 light squares. What does the color count tell you?

Answer: C. It does not rule out a tiling; you still have to find one or show none exists.

Reasoning: Equal counts pass the test, but a region such as region D in this set can still fail.

- ✗ It can certainly be tiled by 6 dominoes.: equal colors are needed, but they are not enough
- ✗ It cannot be tiled by dominoes.: equal colors do not rule out a tiling
- ✗ It can be tiled in exactly 6 ways.: the color count says nothing about the number of ways

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S38-B1-Q5 · question page 878

LOGIC.SPATIAL.05 · Strategy: color the board like a checkerboard

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A region with an even number of squares can always be tiled by dominoes.

Answer: False

Reasoning: The T shape in the worked example has 4 squares and cannot be tiled.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S38-F1-Q1 · question page 879

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a strip 2 squares tall and 6 squares wide be tiled by dominoes?

Answer: 13

Reasoning: Let w(n) be the number for width n. The left end is one upright domino (leaving width n - 1) or two flat ones (leaving n - 2): w(n) = w(n - 1) + w(n - 2). 1, 2, 3, 5, 8, 13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F1-Q2 · question page 880

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row. A 3 by 3 square can be tiled by three straight trominoes in exactly 2 ways.

Answer: True

Reasoning: If the top-left tromino lies flat, the top row is filled and so are the other rows: all flat. If it stands upright, all must stand. 2 ways.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-F1-Q3 · question page 881

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row, flat or upright. One corner square is removed from a 4 by 4 board. Can the other 15 squares be tiled by 5 straight trominoes?

Answer: A. Yes, and there is more than one way.

Reasoning: For example, the row with the gap holds one flat tromino in its other three squares, and the remaining 3 by 4 block can be filled with upright trominoes. A full search finds 4 tilings.

- ✗ No: 15 squares cannot be split into trominoes.: 15 = 5 x 3, so the count is fine
- ✗ No: the corner gap blocks every tiling.: tilings exist; the search finds 4
- ✗ Yes, but in exactly one way.: a search finds 4 different tilings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S38-F1-Q4 · question page 882

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A straight tromino covers three squares in a row, flat or upright. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by straight trominoes?

Answer: 3

Reasoning: All flat does not fit (rows of 4). Each column can hold an upright tromino; a flat block of three flat trominoes fills 3 by 3. Upright x 4; flat block then upright; upright then flat block: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F1-Q5 · question page 883

LOGIC.SPATIAL.05 · Count tilings and use other pieces

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A strip 2 squares tall and n squares wide can be tiled by dominoes in exactly 8 ways. What is n?

Answer: any values that satisfy: n == 5 (example: n = 5)

Reasoning: The counts for widths 1, 2, 3, 4, 5 are 1, 2, 3, 5, 8.

Tempting answers the rule rejects: n = 4; n = 8; n = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Counts by width: [1, 2, 3, 5, 8, 13]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S38-F2-Q1 · question page 884

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. In how many ways can a rectangle 3 squares tall and 4 wide be tiled by dominoes?

Answer: 11

Reasoning: An organized search by the placement covering the top-left square, then the next uncovered square, finds 11 tilings.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S38-F2-Q2 · question page 885

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. A board 4 rows by 5 columns has two squares removed: row 1, column 1 and row 3, column 3. The other 18 squares can be tiled by 9 dominoes.

Answer: False

Reasoning: Both removed squares have the same checkerboard color (1 + 1 and 3 + 3 are even). That leaves 8 of that color and 10 of the other, so no tiling exists.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S38-F2-Q3 · question page 886

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

An L-tromino is three squares in an L shape, turned any way. In how many ways can a strip 2 squares tall and 9 wide be tiled by L-trominoes?

Answer: B. 8

Reasoning: The leftmost L-tromino and its partner always fill a 2 by 3 block, which has 2 tilings. Three blocks: 2 x 2 x 2 = 8.

- ✗ 6: the strip splits into three 2 by 3 blocks with 2 choices each, so multiply, not add
- ✗ 3: that counts the blocks, not the ways
- ✗ 9: that is the width, not a count of tilings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S38-F2-Q4 · question page 887

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a 3 by 3 board. Rows and columns are numbered 1 to 3. Give a square (row r, column c) whose removal leaves 8 squares that can be tiled by dominoes.

Answer: any values that satisfy: isInteger(r) and isInteger(c) and r >= 1 and r <= 3 and c >= 1 and c <= 3 and isEven(r + c) (example: r = 2, c = 2)

Reasoning: The corners and the center are the majority color (5 of 9). Removing one of them leaves 4 of each, and each such board can be tiled.

Tempting answers the rule rejects: r = 1, c = 2; r = 2, c = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Tilings counted for all 9 removals; the 5 with r + c even work. Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S38-F2-Q5 · question page 888

LOGIC.SPATIAL.05 · Prove impossibility and count harder tilings

A domino covers exactly two squares that share a side. A tiling covers every square of the region exactly once, with no domino hanging off the region. One square is removed from a board 3 rows by 5 columns. For how many of the 15 squares does removing it leave 14 squares that can be tiled by dominoes?

Answer: 8

Reasoning: The 8 squares of the majority color (row + column even) are the only candidates, and a search shows each of them works.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-M-Q1 · question page 890

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the steps for solving 4x + 7 = 2x + 19 in order.

Answer: 2x + 7 = 19 (subtract 2x from both sides) → 2x = 12 (subtract 7 from both sides) → x = 6 (divide both sides by 2) → Check: 4(6) + 7 = 31 and 2(6) + 19 = 31

Reasoning: Each line uses the line before it; the check comes last, once x is known.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S39-M-Q2 · question page 891

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Which reason justifies going from 5x - 4 = 21 to 5x = 25?

Answer: D. Add 4 to both sides; equal numbers plus 4 are still equal.

Reasoning: Adding the same number to two equal numbers gives equal results: 21 + 4 = 25.

- ✗ Use the distributive property on 5x - 4.: nothing is distributed in this step
- ✗ Multiply both sides by 4.: multiplying by 4 would give 20x - 16 = 84
- ✗ Rewrite 5x - 4 as 4 - 5x.: subtraction cannot be reordered that way, and the step does not do it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-M-Q3 · question page 892

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student writes: Line 1: 2(x + 3) = 14. Line 2: 2x + 3 = 14. Line 3: 2x = 11. Line 4: x = 5.5. Which line is the first one that does NOT follow from the line before it?

Answer: A. Line 2

Reasoning: 2(x + 3) = 2x + 6, not 2x + 3. The correct steps give 2x = 8 and x = 4.

- ✗ Line 3: from Line 2, subtracting 3 does give 2x = 11
- ✗ Line 4: from Line 3, dividing by 2 does give x = 5.5
- ✗ None; every line follows: Line 2 is wrong, and x = 5.5 does not check: 2(8.5) = 17

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-M-Q4 · question page 893

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. From 6 - x = 10, the next line "x = 10 - 6" follows correctly.

Answer: False

Reasoning: Subtracting 6 gives -x = 4, so x = -4. The line x = 4 does not follow; 6 - 4 = 2, not 10.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-M-Q5 · question page 894

LOGIC.PROOF.02 · Explain each step in solving an equation and justify the solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 7 - 2(x - 1) = x - 3. Give x.

Answer: any values that satisfy: x == 4 (example: x = 4)

Reasoning: 7 - 2x + 2 = x - 3 (distribute), 9 - 2x = x - 3 (combine), 12 = 3x (add 2x and 3), x = 4 (divide by 3). Check: 7 - 2(3) = 1 and 4 - 3 = 1.

Tempting answers the rule rejects: x = 2; x = -4; x = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S39-B1-Q1 · question page 895

LOGIC.PROOF.02 · Properties of equality

If a = b, which statement must also be true?

Answer: C. a - 5 = b - 5

Reasoning: Doing the same thing to two equal numbers keeps them equal.

- ✗ a - 5 = b + 5: different amounts were changed on the two sides
- ✗ 5a = b: only one side was multiplied
- ✗ a / 5 = b / 3: the two sides were divided by different numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-B1-Q2 · question page 896

LOGIC.PROOF.02 · Properties of equality

If 3x = 12, then x = 4 follows by dividing both sides by 3.

Answer: True

Reasoning: 12 / 3 = 4.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-B1-Q3 · question page 897

LOGIC.PROOF.02 · Properties of equality

Which step uses the distributive property?

Answer: B. 5(x - 2) = 13 becomes 5x - 10 = 13

Reasoning: The distributive property rewrites 5(x - 2) as 5x - 10.

- ✗ x + 3 = 8 becomes x = 5: that subtracts 3 from both sides
- ✗ 4x = 20 becomes x = 5: that divides both sides by 4
- ✗ x + 2 = 9 becomes 2 + x = 9: that uses the commutative property

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-B1-Q4 · question page 898

LOGIC.PROOF.02 · Properties of equality

Multiplying both sides of an equation by 0 always gives an equation with the same solutions.

Answer: False

Reasoning: It gives 0 = 0, which every number satisfies, so solutions can be gained.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-B1-Q5 · question page 899

LOGIC.PROOF.02 · Properties of equality

To solve x / 4 = 9 in one step, by what number do you multiply both sides?

Answer: 4

Reasoning: Multiplying both sides by 4 gives x = 36.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-F1-Q1 · question page 900

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. A student solves x squared = 5x by dividing both sides by x and gets x = 5. What is wrong?

Answer: D. Dividing by x assumes x is not 0, and x = 0 is also a solution.

Reasoning: x squared - 5x = 0 gives x(x - 5) = 0, so x = 0 or x = 5.

- ✗ Nothing; x = 5 is the only solution.: x = 0 also works: 0 = 0
- ✗ The answer should be x = 0 only.: x = 5 also works: 25 = 25
- ✗ x squared = 5x has no solutions.: both 0 and 5 are solutions

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F1-Q2 · question page 901

LOGIC.PROOF.02 · Steps that need care

The equation 3(x + 2) = 3x + 6 is true for every number x.

Answer: True

Reasoning: By the distributive property, 3(x + 2) and 3x + 6 are the same expression.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S39-F1-Q3 · question page 902

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 2(x - 4) + 3 = 5x + 7. What is x?

Answer: -4

Reasoning: 2x - 8 + 3 = 5x + 7, so 2x - 5 = 5x + 7, so -12 = 3x, so x = -4. Check: 2(-8) + 3 = -13 and 5(-4) + 7 = -13.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S39-F1-Q4 · question page 903

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solving 4x + 1 = 4x - 3 leads to the line 1 = -3. What does this show?

Answer: C. The equation has no solution.

Reasoning: If some x solved it, then 1 = -3 would follow, which is false. So no x solves it.

- ✗ The solution is x = 0.: x = 0 gives 1 = -3, which is false
- ✗ Every number is a solution.: that would be shown by a true line such as 1 = 1
- ✗ The solution is x = -3.: x = -3 gives -11 = -15, which is false

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S39-F1-Q5 · question page 904

LOGIC.PROOF.02 · Steps that need care

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of (x + 5) / 3 = 4 in order.

Answer: Assume x is a number with (x + 5) / 3 = 4. → x + 5 = 12 (multiply both sides by 3) → x = 7 (subtract 5 from both sides) → Check: (7 + 5) / 3 = 4

Reasoning: Start from the assumption that a solution exists, then undo the division, then the addition, then check.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S39-F2-Q1 · question page 905

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. To solve the equation (square root of (x + 2)) = x, a student squares both sides: x + 2 = x squared, so x = 2 or x = -1. Which answer is correct?

Answer: A. Only x = 2; x = -1 gives square root of 1 = 1, not -1.

Reasoning: Squaring both sides can add solutions, so each candidate must be checked in the original equation.

- ✗ Both x = 2 and x = -1.: x = -1 fails the original equation
- ✗ Only x = -1.: x = -1 fails and x = 2 works: square root of 4 = 2
- ✗ Neither one.: x = 2 works: square root of 4 = 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F2-Q2 · question page 906

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. If each line of a solution follows from the line before it, then every number found at

the end must solve the original equation.

Answer: False

Reasoning: The steps show that any solution is among the numbers found, not that each number found works. Squaring, for example, can add a number that fails the original equation.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S39-F2-Q3 · question page 907

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Solve 3(2x - 1) = 4(x + 2) - 1. Give x.

Answer: any values that satisfy: x == 5 (example: x = 5)

Reasoning: 6x - 3 = 4x + 7, so 2x = 10, so x = 5. Check: 3(9) = 27 and 4(7) - 1 = 27.

Tempting answers the rule rejects: x = 3; x = 2; x = -5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S39-F2-Q4 · question page 908

LOGIC.PROOF.02 · Justify a solution method

Which step can give an equation with MORE solutions than the one before it?

Answer: D. Squaring both sides

Reasoning: Squaring cannot be undone: a = b and a = -b both lead to a squared = b squared.

✗ Adding 7 to both sides: subtracting 7 undoes it, so the solutions stay the same

✗ Dividing both sides by 2: multiplying by 2 undoes it, so the solutions stay the same

✗ Multiplying both sides by -3: dividing by -3 undoes it, so the solutions stay the same

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S39-F2-Q5 · question page 909

LOGIC.PROOF.02 · Justify a solution method

Assume the equation has a solution x. Each new line must follow from the equality in the line before it. Put the lines of this solution of 5 - 3(x - 1) = 2x - 2 in order.

Answer: 5 - 3x + 3 = 2x - 2 (distributive property) → 8 - 3x = 2x - 2 (combine 5 and 3) → 10 = 5x (add 3x and 2 to both sides) → x = 2 (divide both sides by 5)

Reasoning: Distribute, combine like terms, collect x on one side, divide. Check: 5 - 3(1) = 2 and 2(2) - 2 = 2.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-M-Q1 · question page 911

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no largest even integer" by contradiction, how should the proof begin?

Answer: B. Assume there is a largest even integer; call it N.

Reasoning: Assume the opposite. Then N + 2 is even and larger than N,

which is impossible.

- ✗ Assume there is no largest even integer.: that assumes the statement itself, not its opposite
- ✗ Show that 100 is an even integer.: one example says nothing about a largest even integer
- ✗ Assume every even integer is less than 100.: that is a different, narrower assumption

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-M-Q2 · question page 912

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if n squared is even, then n is even, in order.

Answer: Assume n squared is even, but n is odd. → Then n = 2k + 1 for some integer k. → So n squared = 2(2k squared + 2k) + 1, which is odd. → This contradicts n squared being even, so n must be even.

Reasoning: Assume the opposite of the conclusion, use the definition, compute, then point to the contradiction.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-M-Q3 · question page 913

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Reaching a statement that contradicts a known fact shows that the assumption was false.

Answer: True

Reasoning: If the assumption were true, a false statement would follow from it. So the assumption is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-M-Q4 · question page 914

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: no integer is both even and odd. Suppose n = 2a and n = 2b + 1 for integers a and b. Which line gives the contra

Answer: C. 2a = 2b + 1, so 2(a - b) = 1, but 1 is not 2 times an integer.

Reasoning: a - b would be an integer equal to 1/2, which is impossible.

- ✗ 2a = 2b + 1, so a = b.: that algebra is wrong, and it is not a contradiction
- ✗ n = 4 is even and not odd.: one example does not treat a general n
- ✗ 2a + 2b + 1 is an odd number.: adding the two forms is true but contradicts nothing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-M-Q5 · question page 915

LOGIC.PROOF.03 · Set up and complete an argument by contradiction

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To show that 15 is not the sum of two even integers, assume 15 = 2a + 2b for

integers a and b. Then a + b must equal wh

Answer: any values that satisfy: x == 7.5 (example: x = 7.5)

Reasoning: 2(a + b) = 15 gives a + b = 7.5. But a + b is an integer, so the assumption is impossible.

Tempting answers the rule rejects: x = 7; x = 8; x = 15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S40-B1-Q1 · question page 916

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "n is not divisible by 4" by contradiction, what do you assume?

Answer: A. n is divisible by 4.

Reasoning: Assume the exact opposite of the statement.

- ✗ n is not divisible by 4.: that is the statement itself
- ✗ n is divisible by 2.: that is a different statement
- ✗ n is an odd number.: that is a different statement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-B1-Q2 · question page 917

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove "there is no integer x with 2x = 7" by contradiction, what do you assume?

Answer: D. There is an integer x with 2x = 7.

Reasoning: The opposite of "there is no such x" is "there is such an x".

✗ Every integer x has 2x = 7.: the opposite of "no x" is "some x", not "every x"
✗ There is an integer x with 2x not equal to 7.: that is true and is not the opposite
✗ There is no integer x with 2x = 7.: that is the statement itself

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-B1-Q3 · question page 918

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. A proof by contradiction begins by assuming that the statement to be proved is true.

Answer: False

Reasoning: It begins by assuming the statement is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-B1-Q4 · question page 919

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. If assuming P leads to 1 = 0, then P is false.

Answer: True

Reasoning: A true assumption cannot lead to a false statement by valid steps.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-B1-Q5 · question page 920

LOGIC.PROOF.03 · Strategy: assume the opposite and look for the impossible

Definitions: an integer n is even if n = 2k for some integer k, and odd if n = 2k + 1 for some integer k. An integer n is a multiple of m if n = mk for some integer k. In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there is a largest odd integer M. Which number gives the contradiction?

Answer: B. M + 2

Reasoning: M + 2 is odd and larger than M.

- ✗ M + 1: M + 1 is even, so it does not contradict M being the largest odd integer
- ✗ M - 2: M - 2 is smaller than M
- ✗ 2M: 2M is even, so it does not contradict anything

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F1-Q1 · question page 921

LOGIC.PROOF.03 · Contradiction with divisibility

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. To prove that no integers a and b satisfy 6a + 9b = 10, which fact gives the contradiction?

Answer: A. 6a + 9b = 3(2a + 3b) is a multiple of 3, but 10 is not.

Reasoning: If 6a + 9b = 10, then 10 would be 3 times an integer, which it is not.

- ✗ 6 and 9 are both odd numbers.: 6 is even
- ✗ 10 is even, and 6a + 9b is always odd.: a = 1, b = 0 gives 6, which is

even

✗ a and b would have to be fractions.: that is what is to be shown; it is not a contradiction

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F1-Q2 · question page 922

LOGIC.PROOF.03 · Contradiction with divisibility

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of the proof that the square root of 2 is not a fraction in order.

Answer: Assume the square root of 2 = p / q, with p and q integers in lowest terms. → Then p squared = 2q squared, so p squared is even, so p is even: p = 2m. → Then 4m squared = 2q squared, so q squared = 2m squared, and q is even. → Both p and q are even, which contradicts lowest terms.

Reasoning: Each step uses the one before it, and D clashes with the assumption in A.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-F1-Q3 · question page 923

LOGIC.PROOF.03 · Contradiction with divisibility

Among any 5 integers, two of them leave the same remainder when divided by 4.

Answer: True

Reasoning: Assume all 5 remainders are different. But only 4 remainders (0, 1, 2, 3) exist, a contradiction.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S40-F1-Q4 · question page 924

LOGIC.PROOF.03 · Contradiction with divisibility

A perfect square is divided by 4. What is the largest remainder (0, 1, 2 or 3) that can NEVER occur?

Answer: 3

Reasoning: Even squares are (2k) squared = 4k squared (remainder 0); odd squares are 4(k squared + k) + 1 (remainder 1). So 2 and 3 never occur; the larger is 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S40-F1-Q5 · question page 925

LOGIC.PROOF.03 · Contradiction with divisibility

Give an integer n from 1 to 50 for which n squared + 1 is a multiple of 3. If no such n exists, enter 0.

Answer: any values that satisfy: n == 0 (example: n = 0)

Reasoning: Assume some n works. n leaves remainder 0, 1 or 2 when divided by 3, so n squared leaves 0, 1 or 1; then n squared + 1 leaves 1 or 2, never 0. Contradiction, so none exists.

Tempting answers the rule rejects: n = 1; n = 2; n = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: no n from 1 to 50 works. Automatic: the example passes the rule and 3 listed mistakes fail it.

S40-F2-Q1 · question page 926

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Assume there are only finitely many primes, p1, p2, ..., pk, and let N = p1 x p2 x ... x pk + 1. Why is this a contradiction?

Answer: D. N leaves remainder 1 when divided by each listed prime, so a prime factor of N is not on the list.

Reasoning: Every integer greater than 1 has a prime factor, and none of the listed primes divides N.

- ✗ N is always a prime number.: not always: 2 x 3 x 5 x 7 x 11 x 13 + 1 = 30031 = 59 x 509
- ✗ N is always an even number.: N is one more than an even product, so it is odd
- ✗ N is larger than every prime, so N is not prime.: being large does not make a number composite

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F2-Q2 · question page 927

LOGIC.PROOF.03 · Contradiction in larger arguments

2 x 3 x 5 x 7 x 11 x 13 + 1 is a prime number.

Answer: False

Reasoning: 30031 = 59 x 509.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S40-F2-Q3 · question page 928

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Put the steps of a proof that if a + b is at least 20, then a is at least 10 or b is at least 10, in order.

Answer: Assume a + b is at least 20, but a < 10 and b < 10. → Adding the two inequalities gives a + b < 20. → That contradicts a + b being at least 20. → So a is at least 10 or b is at least 10.

Reasoning: The opposite of "a at least 10 or b at least 10" is "a < 10 and b < 10".

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 1 valid.

S40-F2-Q4 · question page 929

LOGIC.PROOF.03 · Contradiction in larger arguments

In a proof by contradiction, you assume the statement is false and show that this leads to something impossible. Claim: every integer is even. "Proof: Assume some integer is odd, say 3. Then 3 = 2k for no integer k. Contradiction." What is wrong?

Answer: B. Nothing impossible was reached; 3 not being 2k agrees with 3 being odd.

Reasoning: A contradiction must clash with the assumption or a known fact. Here everything found is consistent.

- ✗ It should assume that every integer is odd.: the opposite of "every integer is even" is "some integer is not even"
- ✗ It should use 5 instead of 3.: any odd example has the same problem
- ✗ Nothing; the proof is correct.: the claim is false, since 3 is odd

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S40-F2-Q5 · question page 930

LOGIC.PROOF.03 · Contradiction in larger arguments

At least how many integers must you choose to be sure that two of them leave the same remainder when divided by 6?

Answer: 7

Reasoning: Six integers can have the six different remainders 0 to 5. With 7, assuming all remainders differ needs 7 remainders, but only 6 exist.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-M-Q1 · question page 932

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Seven gears form a closed ring, each touching the gears on either side (figure). Gear 1 is turned clockwise. What happens?

Answer: C. The gears jam: going around the ring, gear 1 would have to turn both ways.

Reasoning: Going around: 1 clockwise, 2 counter, 3 clockwise, ..., 7 clockwise. Gear 7 touches gear 1, so they must turn opposite ways, but both are clockwise. An odd ring cannot turn.

- ✗ Every gear turns, and the directions alternate around the ring.: alternating around 7 gears ends with gear 7 clockwise, touching gear 1, which is also clockwise
- ✗ Every gear turns clockwise.: touching gears must turn opposite ways
- ✗ Only gears 1, 3, 5 and 7 turn.: gears that touch a turning gear are forced to turn

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-M-Q2 · question page 933

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Twelve gears stand in a row (not a ring), each touching the next. Gear 1 turns clockwise. How many of the 12 gears turn counterclockwise?

Answer: 6

Reasoning: Odd-numbered gears turn clockwise and even-numbered gears counterclockwise: 2, 4, ..., 12 is 6 gears.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-M-Q3 · question page 934

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A frog starts on 0 on a number line and makes exactly 9 jumps, each 1 unit left or right. It can end on 4.

Answer: False

Reasoning: Each jump changes the position by 1, so it switches between even and odd. After 9 jumps (odd) the frog is on an odd number, and 4 is even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-M-Q4 · question page 935

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A lamp starts OFF. Each press of its switch changes it (OFF to ON, or ON to OFF). Give a number of presses from 1 to 20 that leaves the lamp ON.

Answer: any values that satisfy: isInteger(n) and n >= 1 and n <= 20 and isOdd(n) (example: n = 7)

Reasoning: After an odd number of presses the lamp is ON; after an even number it is OFF.

Tempting answers the rule rejects: n = 8; n = 20; n = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 10 valid answers.

S41-M-Q5 · question page 936

LOGIC.INVARIANT.01 · Use odd/even and alternation to predict outcomes and rule out possibilities

A token moves along a row of squares numbered 1 to 15. Each move jumps exactly 2 squares left or right (staying on the row). It starts on square 4. How many of the 15 squares can it ever reach, counting square 4?

Answer: 7

Reasoning: Jumps of 2 keep the token on even squares: 2, 4, 6, 8, 10, 12, 14. That is 7 squares, and all of them can be reached.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-B1-Q1 · question page 937

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

The sum of 5 odd numbers is odd.

Answer: True

Reasoning: Pairs of odd numbers add to even numbers; 5 odd numbers make 2 pairs and 1 left over, so the sum is even + odd = odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-B1-Q2 · question page 938

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

A coin starts heads up. It is turned over 13 times. Which side is up at the end?

Answer: B. Tails

Reasoning: 13 is odd, so the side switches an odd number of times: tails.

- ✗ Heads: heads is up only after an even number of turns
- ✗ Either side, depending on how it is turned: every turn switches the side, so the result is fixed
- ✗ It cannot be known without trying: the parity of 13 decides it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S41-B1-Q3 · question page 939

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears stand in a row, each touching the next. Gear 1 turns clockwise. What is the number of the last gear in the row that turns clockwise?

Answer: 9

Reasoning: Odd-numbered gears turn clockwise, and 9 is odd.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-B1-Q4 · question page 940

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

An even number plus an odd number is even.

Answer: False

Reasoning: 2 + 3 = 5. Even + odd is always odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S41-B1-Q5 · question page 941

LOGIC.INVARIANT.01 · Strategy: keep track of odd and even only

Which total CANNOT be the sum of three odd numbers?

Answer: C. 10

Reasoning: odd + odd + odd = even + odd = odd, so the total is always odd.

✗ 9: 3 + 3 + 3 = 9
✗ 15: 5 + 5 + 5 = 15
✗ 3: 1 + 1 + 1 = 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F1-Q1 · question page 942

LOGIC.INVARIANT.01 · Parity on boards and rings

A token starts on a corner square of a 5 by 5 board. Each move goes to a square sharing a side (going back is allowed). After exactly 7 moves, how many different squares could the token be on?

Answer: 12

Reasoning: Color the board like a checkerboard. Each move changes color, so after 7 moves the token is on the other color from the corner: the 12 such squares. All 12 are within 7 moves.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F1-Q2 · question page 943

LOGIC.INVARIANT.01 · Parity on boards and rings

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Ten gears form a closed ring, each touching the gears on either side. They can all turn.

Answer: True

Reasoning: Alternating around 10 gears ends with gear 10 counterclockwise, opposite to gear 1, which it touches. No conflict.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S41-F1-Q3 · question page 944

LOGIC.INVARIANT.01 · Parity on boards and rings

Seven cups stand upside down. A move turns over exactly 4 cups (any 4). Can all seven cups end right side up?

Answer: A. No: each move changes the number of upside-down cups by an even amount, so it stays odd.

Reasoning: Turning 4 cups changes the upside-down count by -4, -2, 0, 2 or 4. Starting from 7 (odd), it stays odd and can never be 0.

- ✗ Yes: turn over 4 cups, then 3 cups.: a move must turn exactly 4 cups
- ✗ Yes, after exactly 7 moves.: no number of moves works
- ✗ No: 7 is not a multiple of 4.: the true reason is parity; for example 6 cups could all be turned right side up

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F1-Q4 · question page 945

LOGIC.INVARIANT.01 · Parity on boards and rings

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. Give a total from 0 to 10 that can be made this way.

Answer: any values that satisfy: isInteger(t) and t >= 0 and t <= 10 and isOdd(t) (example: t = 1)

Reasoning: 1 + 2 + ... + 10 = 55 is odd. Changing a + to a - lowers the total by an even amount, so every total is odd. Each odd total from 1 to 9 can be made.

Tempting answers the rule rejects: t = 0; t = 4; t = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: the totals from 0 to 10 that can be made are [1, 3, 5, 7, 9]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S41-F1-Q5 · question page 946

LOGIC.INVARIANT.01 · Parity on boards and rings

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Nine gears are arranged in a 3 by 3 square, and each gear touches the

gears directly beside, above and below it. A corner gear turns clockwise. How many gears turn clockwise, including the corner gear?

Answer: 5

Reasoning: Color like a checkerboard: the corners and center share the corner's color (5 gears) and turn clockwise; the other 4 turn counterclockwise.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F2-Q1 · question page 947

LOGIC.INVARIANT.01 · Prove impossibility with parity

A token must visit every square of a 5 by 5 board exactly once, each move going to a square that shares a side. It starts on the square next to a corner (row 1, column 2). Is this possible?

Answer: D. No: the path must start and end on the corner color, and this square has the other color.

Reasoning: With checkerboard colors there are 13 corner-colored and 12 other squares. A path alternates colors, so a 25-square path must have 13 of one color at positions 1, 3, ..., 25: it starts on the corner color. A full search confirms no path starts at row 1, column 2.

- ✗ Yes: go along row 1, then snake back and forth.: that path misses the corner square or repeats one
- ✗ Yes, but only if it ends on a corner.: no such path exists from this square at all
- ✗ No: 25 is not a multiple of 5.: 25 = 5 x 5; the count is not the problem

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S41-F2-Q2 · question page 948

LOGIC.INVARIANT.01 · Prove impossibility with parity

Put a + or - sign in front of each of the numbers 1, 2, 3, ..., 10 and add. How many different totals can be made?

Answer: 56

Reasoning: Totals are odd, from -55 to 55. Every odd number in that range can be made: 56 totals.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S41-F2-Q3 · question page 949

LOGIC.INVARIANT.01 · Prove impossibility with parity

On a square grid, a path that moves one square up, down, left or right at each step, and ends where it started, always has an even number of steps.

Answer: True

Reasoning: Each step changes the checkerboard color. To return to the starting color, the number of changes must be even.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S41-F2-Q4 · question page 950

LOGIC.INVARIANT.01 · Prove impossibility with parity

Gears that touch mesh: when one turns clockwise, a gear touching it turns counterclockwise, and the reverse. Give a number of gears n from 3 to 12 for which a closed ring of n gears, each touching its two neighbors, would jam.

Answer: any values that satisfy: isInteger(n) and n >= 3 and n <= 12 and

isOdd(n) (example: n = 5)

Reasoning: A ring jams exactly when the number of gears is odd: 3, 5, 7, 9 or 11.

Tempting answers the rule rejects: n = 4; n = 12; n = 13

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 5 valid answers.

S41-F2-Q5 · question page 951

LOGIC.INVARIANT.01 · Prove impossibility with parity

A bug walks along the edges of a cube from corner to corner. It starts at one corner and makes exactly 5 moves (going back is allowed). How many of the 8 corners could it end on?

Answer: 4

Reasoning: Color the corners in two colors so every edge joins different colors (4 and 4). After an odd number of moves the bug is on the other color: 4 corners, and each can be reached.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-M-Q1 · question page 953

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: for every whole number n, n squared + n + 11 is prime. Which n is a counterexample?

Answer: B. 10

Reasoning: 100 + 10 + 11 = 121 = 11 x 11, which is not prime.

✗ 1: 1 + 1 + 11 = 13, which is prime

✗ 2: 4 + 2 + 11 = 17, which is prime

✗ 3: 9 + 3 + 11 = 23, which is prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-M-Q2 · question page 954

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: for all numbers a and b, if a > b, then a squared > b squared. Give a counterexample.

Answer: any values that satisfy: a > b and a^2 <= b^2 (example: a = 1, b = -2)

Reasoning: For example a = 1, b = -2: 1 > -2, but 1 is not greater than 4. A negative b works.

Tempting answers the rule rejects: a = 3, b = 2; a = 2, b = 3; a = 5, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S42-M-Q3 · question page 955

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. The claim "every odd number greater than 1 is prime" is disproved by the number 9.

Answer: True

Reasoning: 9 is odd and greater than 1, but 9 = 3 x 3.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-M-Q4 · question page 956

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, plus 1, is prime for every whole number n from 1 to 6. What is the smallest n that disproves it?

Answer: 3

Reasoning: n = 1: 3, prime. n = 2: 5, prime. n = 3: 9 = 3 x 3, not prime.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-M-Q5 · question page 957

LOGIC.PROOF.04 · Find a counterexample that disproves a general mathematical claim

A counterexample to a claim about every case is one case where the claim is false. Claim: every multiple of 4 is a multiple of 8. Which number is NOT a counterexample?

Answer: A. 16

Reasoning: 16 is a multiple of 8, so the claim holds for 16.

- ✗ 4: 4 is a multiple of 4 but not of 8, so it is a counterexample
- ✗ 12: 12 is a multiple of 4 but not of 8, so it is a counterexample
- ✗ 20: 20 is a multiple of 4 but not of 8, so it is a counterexample

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-B1-Q1 · question page 958

LOGIC.PROOF.04 · What a counterexample is

What is a counterexample to the claim "every number with property P has property Q"?

Answer: D. A number with property P that does not have property Q

Reasoning: The claim fails exactly when P holds and Q does not.

- ✗ A number with property Q that does not have property P: that does not test the claim
- ✗ A number with neither property: the claim says nothing about numbers without P
- ✗ A number with both properties: that agrees with the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S42-B1-Q2 · question page 959

LOGIC.PROOF.04 · What a counterexample is

One counterexample is enough to show that a claim about every case is false.

Answer: True

Reasoning: The claim said every case works; one failing case shows it does not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S42-B1-Q3 · question page 960

LOGIC.PROOF.04 · What a counterexample is

If a claim works for five examples, it is proved true for every case.

Answer: False

Reasoning: A later case could still fail.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S42-B1-Q4 · question page 961

LOGIC.PROOF.04 · What a counterexample is

A counterexample to a claim about every case is one case where the claim is false. Which number is a counterexample to "all prime numbers are odd"?

Answer: C. 2

Reasoning: 2 is prime and even.

- ✗ 9: 9 is odd and not prime, so it does not test the claim
- ✗ 3: 3 is prime and odd, so it agrees with the claim
- ✗ 4: 4 is not prime

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-B1-Q5 · question page 962

LOGIC.PROOF.04 · What a counterexample is

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number divisible by 3 is divisible by 6. Give a counterexample from 1 to 30.

Answer: any values that satisfy: isInteger(n) and n >= 1 and n <= 30 and mod(n, 3) == 0 and mod(n, 6) != 0 (example: n = 9)

Reasoning: Any odd multiple of 3 works: 3, 9, 15, 21, 27.

Tempting answers the rule rejects: n = 6; n = 10; n = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the

stated domain: 5 valid answers.

S42-F1-Q1 · question page 963

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: (a + b) squared = a squared + b squared for all numbers a and b. Give whole numbers a and b that disprove it.

Answer: any values that satisfy: (a + b)^2 != a^2 + b^2 (example: a = 1, b = 1)

Reasoning: For a = 1, b = 1: 4 is not 2. The claim fails whenever a and b are both nonzero.

Tempting answers the rule rejects: a = 0, b = 5; a = 3, b = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S42-F1-Q2 · question page 964

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: the square root of (a + b) equals the square root of a plus the square root of b, for all numbers a and b that are 0 or more. Which pair is a counterexample?

Answer: B. a = 9, b = 16

Reasoning: Square root of 25 is 5, but 3 + 4 = 7.

- ✗ a = 0, b = 4: square root of 4 = 0 + 2, so the claim holds here
- ✗ a = 0, b = 0: 0 = 0 + 0, so the claim holds here
- ✗ a = 1, b = 0: 1 = 1 + 0, so the claim holds here

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the

check function accepts only the keyed choice.

S42-F1-Q3 · question page 965

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: n factorial plus 1 is prime for every positive whole number n. (n factorial is 1 x 2 x ... x n.) What is the smallest n that disproves it?

Answer: 4

Reasoning: 1! + 1 = 2, 2! + 1 = 3, 3! + 1 = 7 are prime; 4! + 1 = 25 = 5 x 5 is not.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-F1-Q4 · question page 966

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. The claim "if n squared is a multiple of 4, then n is a multiple of 4" is disproved by n = 2.

Answer: True

Reasoning: 2 squared = 4 is a multiple of 4, but 2 is not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-F1-Q5 · question page 967

LOGIC.PROOF.04 · Counterexamples to algebra rules

A counterexample to a claim about every case is one case where the claim is false. Claim: every rectangle with perimeter 20 has area 25. Which rectangle is a counterexample?

Answer: D. 4 by 6

Reasoning: A 4 by 6 rectangle has perimeter 20 but area 24.

- ✗ 5 by 5: perimeter 20 and area 25, so it agrees with the claim
- ✗ 2 by 10: its perimeter is 24, so the claim says nothing about it
- ✗ 3 by 8: its perimeter is 22, so the claim says nothing about it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S42-F2-Q1 · question page 968

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for whole numbers a, b and c, if a divides b x c, then a divides b or a divides c. Give a counterexample with a, b and c each from 2 to 12.

Answer: any values that satisfy: isInteger(a) and isInteger(b) and isInteger(c) and a >= 2 and a <= 12 and b >= 2 and b <= 12 and c >= 2 and c <= 12 and mod(b * c, a) == 0 and mod(b, a) != 0 and mod(c, a) != 0 (example: a = 4, b = 2, c = 6)

Reasoning: For example a = 4, b = 2, c = 6: 4 divides 12, but 4 divides neither 2 nor 6. (The claim is true when a is prime.)

Tempting answers the rule rejects: a = 3, b = 3, c = 5; a = 5, b = 2, c = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it. Search of the stated domain: 77 valid answers.

S42-F2-Q2 · question page 969

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: 2 to the power n, minus 1, is prime whenever n is prime. The smallest counterexample is n = 11.

Answer: True

Reasoning: n = 2, 3, 5, 7 give 3, 7, 31, 127, all prime. n = 11 gives 2047 = 23 x 89.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S42-F2-Q3 · question page 970

LOGIC.PROOF.04 · Counterexamples that take a search

Which claim is FALSE?

Answer: C. If n is prime, then n + 2 is prime.

Reasoning: Counterexample: 7 is prime, but 9 is not.

- ✗ If n is odd, then n squared is odd.: true: (2k + 1) squared = 2(2k squared + 2k) + 1
- ✗ The sum of two even numbers is even.: true: 2a + 2b = 2(a + b)
- ✗ If n squared is odd, then n is odd.: true: if n were even, n squared would be even

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S42-F2-Q4 · question page 971

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: every whole number from 1 to 30 is a sum of two square numbers (0, 1, 4, 9, 16, 25, ...; a square may be used twice). What is the smallest counterexample?

Answer: 3

Reasoning: 1 = 1 + 0, 2 = 1 + 1, but 3 cannot be made: the squares up to 3 are 0 and 1.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S42-F2-Q5 · question page 972

LOGIC.PROOF.04 · Counterexamples that take a search

A counterexample to a claim about every case is one case where the claim is false. Claim: for every positive whole number n, 2 to the power n is greater than n. Why is n = -1 NOT a valid counterexample?

Answer: A. The claim is only about positive whole numbers, and -1 is not one.

Reasoning: A counterexample must be a case the claim is about.

- ✗ Because 2 to the power -1 is greater than -1.: that is true, but it is not why -1 cannot count
- ✗ Because a counterexample must be a large number.: any case covered by the claim may serve
- ✗ Because counterexamples cannot be negative.: they can be, when the claim covers negative numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-M-Q1 · question page 974

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

Lines meet at a point, forming angle 1 and angle 2 as vertical angles, and angle 3 next to both (angle 3 forms a linear pair with angle 1 and with angle 2). Put the steps of the proof that angle 1 = angle 2 in a correct order. (Two orders are correct.)

Answer: angle 1 + angle 3 = 180 degrees (linear pair) → angle 2 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees) → angle 1 = angle 2 (subtract angle 3

from both sides)

Also accepted: angle 2 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = 180 degrees (linear pair) → angle 1 + angle 3 = angle 2 + angle 3 (both sums equal 180 degrees) → angle 1 = angle 2 (subtract angle 3 from both sides)

Reasoning: Both linear pairs come first (in either order), then substitution, then subtraction.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-M-Q2 · question page 975

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Which pair of angles are alternate interior angles?

Answer: B. Angles 4 and 5

Reasoning: Angles 4 and 5 lie between p and q, on opposite sides of t.

- ✗ Angles 1 and 5: they are corresponding angles: same position at each line
- ✗ Angles 2 and 3: they are vertical angles at line p
- ✗ Angles 4 and 8: they are corresponding angles: same position at each line

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-M-Q3 · question page 976

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by

intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. If angle 2 = 70 degrees, then angle 7 = 70 degrees.

Answer: True

Reasoning: Angle 2 = angle 3 (vertical), angle 3 = angle 6 (alternate interior, p parallel to q), and angle 6 = angle 7 (vertical). So angle 7 = 70 degrees.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S43-M-Q4 · question page 977

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel, and angle 1 = 115 degrees. Give the measure of angle 6 in degrees.

Answer: any values that satisfy: x == 65 (example: x = 65)

Reasoning: Angle 1 = angle 5 (corresponding), and angle 5 + angle 6 = 180 degrees (linear pair), so angle 6 = 65 degrees.

Tempting answers the rule rejects: x = 115; x = 75; x = 245

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S43-M-Q5 · question page 978

LOGIC.PROOF.05 · Order and justify the steps of proofs about angles formed by

intersecting and parallel lines

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 3 = angle 6 when p is parallel to q in a correct order. (Two orders are correct.)

Answer: p is parallel to q, cut by transversal t (given) → angle 2 = angle 6 (corresponding angles, p parallel to q) → angle 2 = angle 3 (vertical angles) → angle 3 = angle 6 (both equal angle 2)

Also accepted: p is parallel to q, cut by transversal t (given) → angle 2 = angle 3 (vertical angles) → angle 2 = angle 6 (corresponding angles, p parallel to q) → angle 3 = angle 6 (both equal angle 2)

Reasoning: Start with the given. The corresponding-angles step and the vertical-angles step can come in either order; the conclusion comes last.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-B1-Q1 · question page 979

LOGIC.PROOF.05 · Name the angle pairs

Two angles form a linear pair (together they make a straight line). One measures 48 degrees. What does the other measure, in degrees?

Answer: 132

Reasoning: 180 - 48 = 132.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S43-B1-Q2 · question page 980

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 1 and 4 are vertical angles.

Answer: True

Reasoning: They are opposite each other at the same crossing.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-B1-Q3 · question page 981

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). What kind of angle pair are angles 3 and 7?

Answer: C. Corresponding angles

Reasoning: Both are in the lower-left position at their crossings.

- ✗ Vertical angles: vertical angles are at the same crossing
- ✗ Alternate interior angles: angle 7 is below q, outside the two lines
- ✗ A linear pair: a linear pair is at one crossing and shares a side

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-B1-Q4 · question page 982

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angles 3 and 5 are alternate interior angles.

Answer: False

Reasoning: Both are on the left of t; they are same-side interior angles. The alternate interior pairs are 3 and 6, and 4 and 5.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-B1-Q5 · question page 983

LOGIC.PROOF.05 · Name the angle pairs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Angle 3 = 2x + 10 degrees and angle 2 = 3x - 15 degrees. They are vertical angles. Find x.

Answer: any values that satisfy: x == 25 (example: x = 25)

Reasoning: Vertical angles are equal: 2x + 10 = 3x - 15, so x = 25.

Tempting answers the rule rejects: x = 5; x = 37

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F1-Q1 · question page 984

LOGIC.PROOF.05 · Perpendicular bisectors and converses

M is the midpoint of segment AB, and line PM is perpendicular to AB, so P is on the perpendicular bisector of AB. Put the steps of the proof that PA = PB in a correct order. (Two orders are correct.)

Answer: AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular) → PM = PM (a segment equals itself) → Triangle PMA is congruent to triangle PMB (SAS) → PA = PB (corresponding parts of congruent triangles)

Also accepted: PM = PM (a segment equals itself) → AM = BM, and angles PMA and PMB are right angles (midpoint; perpendicular) →

Triangle PMA is congruent to triangle PMB (SAS) → PA = PB (corresponding parts of congruent triangles)

Reasoning: The two side facts and the angle fact come first, in either order; then SAS, then the conclusion.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 24 orderings of 4 items enumerated; 2 valid.

S43-F1-Q2 · question page 985

LOGIC.PROOF.05 · Perpendicular bisectors and converses

Theorem: a point is on the perpendicular bisector of AB exactly when it is the same distance from A and B. Point P has PA = PB = 7. What follows?

Answer: D. P is on the perpendicular bisector of AB.

Reasoning: The "exactly when" part means equal distances put P on the perpendicular bisector.

- ✗ P is the midpoint of AB.: P need not be on segment AB at all
- ✗ AB = 7.: the distances to A and B do not fix the length of AB
- ✗ Nothing follows without more information.: the theorem works in both directions

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F1-Q3 · question page 986

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Then angle 4 + angle 6 = 180 degrees.

Answer: True

Reasoning: Angle 4 = angle 5 (alternate interior), and angle 5 + angle 6 = 180 degrees (linear pair).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S43-F1-Q4 · question page 987

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 4 = 3x + 5 degrees and angle 5 = 5x - 35 degrees. Find x.

Answer: any values that satisfy: x == 20 (example: x = 20)

Reasoning: Alternate interior angles are equal: 3x + 5 = 5x - 35, so x = 20 (both angles are 65 degrees).

Tempting answers the rule rejects: x = 15; x = 65

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F1-Q5 · question page 988

LOGIC.PROOF.05 · Perpendicular bisectors and converses

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Here p and q are NOT known to be parallel. Which fact would prove that they are parallel?

Answer: A. angle 3 = angle 6

Reasoning: If alternate interior angles are equal, the lines are parallel (the converse theorem).

✗ angle 1 = angle 4: vertical angles are always equal, parallel or not

✗ angle 1 + angle 2 = 180 degrees: a linear pair always adds to 180 degrees

✗ angle 5 = angle 8: vertical angles are always equal, parallel or not

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F2-Q1 · question page 989

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Put the steps of the proof that angle 4 + angle 6 = 180 degrees when p is parallel to q in a correct order. (Two orders are correct.)

Answer: angle 4 = angle 5 (alternate interior angles, p parallel to q) → angle 5 + angle 6 = 180 degrees (linear pair) → angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

Also accepted: angle 5 + angle 6 = 180 degrees (linear pair) → angle 4 = angle 5 (alternate interior angles, p parallel to q) → angle 4 + angle 6 = 180 degrees (substitute angle 4 for angle 5)

Reasoning: A and B in either order, then the substitution.

Grading: The learner's order must equal the key or an accepted alternative.

Verified: Automatic: all 6 orderings of 3 items enumerated; 2 valid.

S43-F2-Q2 · question page 990

LOGIC.PROOF.05 · Complete and check angle proofs

A student writes: "Angle 1 = angle 4 because they look the same size in the diagram." What is wrong with this step?

Answer: A. A diagram is not a reason; the step needs the vertical angles theorem.

Reasoning: Each step of a proof must cite a definition, given or theorem.

- ✗ Angles 1 and 4 are never equal.: vertical angles are always equal
- ✗ The step should say angle 1 + angle 4 = 180 degrees.: that is false for vertical angles in general
- ✗ Nothing; looking at the diagram is enough.: diagrams can be drawn inaccurately

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S43-F2-Q3 · question page 991

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Lines p and q are parallel. Angle 1 = 2x + 30 degrees and angle 6 = x + 15 degrees. Find x.

Answer: any values that satisfy: x == 45 (example: x = 45)

Reasoning: Angle 1 = angle 5 (corresponding), and angle 5 + angle 6 = 180. So 2x + 30 + x + 15 = 180, 3x = 135, x = 45.

Tempting answers the rule rejects: x = 15; x = -15

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 2 listed mistakes fail it.

S43-F2-Q4 · question page 992

LOGIC.PROOF.05 · Complete and check angle proofs

In the figure, lines p and q are cut by the transversal t. Angles 1 to 4 are at line p and angles 5 to 8 at line q, in the same positions: 1 upper left, 2 upper right, 3 lower left, 4 lower right (and 5, 6, 7, 8 likewise). Without knowing that p and q are parallel, one can conclude that angle 3 = angle 6.

Answer: False

Reasoning: Alternate interior angles are equal only when the lines are parallel. If p is tilted, angle 3 and angle 6 differ.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S43-F2-Q5 · question page 993

LOGIC.PROOF.05 · Complete and check angle proofs

P is on the perpendicular bisector of segment AB. PA = 3x - 4 and PB = x + 10. What is PA?

Answer: 17

Reasoning: Points on the perpendicular bisector are equidistant from A and B: 3x - 4 = x + 10, so x = 7 and PA = 17.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-M-Q1 · question page 995

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

The numbers 1, 2, 3, ..., 10 are written on a board. A move erases any two numbers and writes their sum. After 9 moves one number is left. What is it?

Answer: B. 55

Reasoning: Erasing a and b and writing a + b does not change the total. The total starts at 55, so the last number is 55.

- ✗ 10: the moves combine numbers; they do not count them
- ✗ 45: that is the sum of 1 to 9; the 10 is still there
- ✗ It depends on the moves chosen: every move keeps the total the

same

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-M-Q2 · question page 996

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

The numbers 1, 2, ..., 10 are written on a board. A move erases two numbers and writes their difference (larger minus smaller). After 9 moves one number is left. It can be 0.

Answer: False

Reasoning: Replacing a and b by a - b changes the total by 2b, an even amount, so the total stays odd (it starts at 55). The last number is odd, so it cannot be 0.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-M-Q3 · question page 997

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Seven coins lie in a row: H H T H T T T (H = heads, T = tails). A move turns over any two coins that are next to each other. What is the smallest number of heads that can ever be showing?

Answer: 1

Reasoning: A move changes the number of heads by -2, 0 or +2, so it stays odd (it starts at 3). It can reach 1 but never 0.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-M-Q4 · question page 998

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

Seven coins lie in a row: H H T H T T T. A move turns over any two coins that are next to each other. Give a number of heads (0 to 7) that can be showing after some moves.

Answer: any values that satisfy: isInteger(h) and h >= 0 and h <= 7 and isOdd(h) (example: h = 5)

Reasoning: The number of heads stays odd: 1, 3, 5 or 7 are possible, and all four can be reached.

Tempting answers the rule rejects: h = 0; h = 4; h = 2

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Breadth-first search over all reachable rows: head counts [1, 3, 5, 7]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S44-M-Q5 · question page 999

LOGIC.INVARIANT.02 · Find a quantity that never changes under the allowed moves and use it to prove a target is impossible

A robot starts at the point (0, 0). Each move takes it from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). Which point can it reach?

Answer: C. (4, 5)

Reasoning: x + y is always a multiple of 3. (4, 5): two moves of (1, 2) and one of (2, 1).

- ✗ (5, 5): each move adds 3 to x + y, but 5 + 5 = 10 is not a multiple of 3
- ✗ (2, 3): 2 + 3 = 5 is not a multiple of 3
- ✗ (7, 0): 7 + 0 = 7 is not a multiple of 3, and y can never stay 0

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-B1-Q1 · question page 1000

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A counter shows 12. A move adds 4 or subtracts 6. Can it ever show 15?

Answer: D. No: every move keeps the number even, and 15 is odd.

Reasoning: Adding 4 or subtracting 6 keeps an even number even.

- ✗ Yes: add 4, then subtract 1.: subtracting 1 is not an allowed move
- ✗ Yes: after enough moves.: no number of moves can change even to odd
- ✗ No: 15 is not a multiple of 4.: 12 + 4 + 4 - 6 = 14 is reachable and is not a multiple of 4 either; the true reason is parity

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S44-B1-Q2 · question page 1001

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

Three numbers on a board are 2, 4 and 9. A move adds 2 to any one of them. The total of the three numbers always stays odd.

Answer: True

Reasoning: The total starts at 15 (odd), and each move adds 2, which keeps it odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S44-B1-Q3 · question page 1002

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A move erases two numbers a and b on a board and writes a + b. The board starts with 3, 5, 6 and 8. What number is left after 3 moves?

Answer: 22

Reasoning: The total never changes: 3 + 5 + 6 + 8 = 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-B1-Q4 · question page 1003

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

If a quantity is the same after every allowed move, it can never change, however many moves are made.

Answer: True

Reasoning: Each move leaves it unchanged, so a chain of moves does too.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S44-B1-Q5 · question page 1004

LOGIC.INVARIANT.02 · Strategy: look for something that never changes

A robot starts at (0, 0). Each move goes from (x, y) to (x + 2, y) or to (x, y + 2). Which point can it NEVER reach?

Answer: B. (3, 4)

Reasoning: Both coordinates always stay even, and 3 is odd.

✗ (2, 4): reached by (2, 0), then (2, 2), then (2, 4)

✗ (0, 6): three moves of (0, 2)

✗ (4, 0): two moves of (2, 0)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-F1-Q1 · question page 1005

LOGIC.INVARIANT.02 · Invariants with remainders

The numbers 1, 2, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. After 9 moves one number is left. What is it?

Answer: 64

Reasoning: Each move raises the total by exactly 1. The total starts at 55, and 9 moves add 9: 64.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F1-Q2 · question page 1006

LOGIC.INVARIANT.02 · Invariants with remainders

A robot starts at (0, 0). Each move goes from (x, y) to (x + 1, y + 2) or to (x + 2, y + 1). It can reach (10, 10).

Answer: False

Reasoning: x + y must be a multiple of 3, and 20 is not.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-F1-Q3 · question page 1007

LOGIC.INVARIANT.02 · Invariants with remainders

Three numbers are on a board: 1, 2, 3. A move picks two of them, adds 1 to one and subtracts 1 from the other. Which quantity never changes?

Answer: C. The total of the three numbers

Reasoning: The move adds 1 and subtracts 1, so the total is always 6.

- ✗ The largest number: adding 1 to the largest number changes it
- ✗ The product of the three numbers: 1, 2, 3 can become 0, 3, 3, whose product is 0
- ✗ The number of odd numbers: 1, 2, 3 can become 2, 2, 2, which has no odd numbers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S44-F1-Q4 · question page 1008

LOGIC.INVARIANT.02 · Invariants with remainders

Ten coins all show heads. A move turns over any four coins (they need not be next to each other). Give a number of heads from 0 to 10 that can NEVER be showing.

Answer: any values that satisfy: isInteger(h) and h >= 0 and h <= 10 and isOdd(h) (example: h = 7)

Reasoning: If k of the four turned coins were heads, the heads count changes by (4 - k) - k = 4 - 2k, an even number. It starts at 10, so it stays even: 1, 3, 5, 7 and 9 are impossible.

Tempting answers the rule rejects: h = 0; h = 6; h = 10

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search of all reachable head counts: [0, 2, 4, 6, 8, 10]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S44-F1-Q5 · question page 1009

LOGIC.INVARIANT.02 · Invariants with remainders

A counter shows 100. A move subtracts 7 or adds 3. What is the smallest number of moves that makes it show 0?

Answer: 20

Reasoning: If a moves subtract 7 and b moves add 3, then 7a - 3b = 100. So 7a is at least 100 and 7a - 100 is a multiple of 3: a = 16 gives 112 - 100 = 12, so b = 4. Total 16 + 4 = 20, and any other solution has larger a and b.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F2-Q1 · question page 1010

LOGIC.INVARIANT.02 · Prove targets impossible

The numbers 1, 2, 3, 4 and 5 are on a board. A move erases two numbers a and b and writes a x b + a + b. After 4 moves one number is left. What is it?

Answer: 719

Reasoning: a x b + a + b = (a + 1)(b + 1) - 1. So the product of (each number + 1) never changes: 2 x 3 x 4 x 5 x 6 = 720. The last number n has n + 1 = 720, so n = 719.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S44-F2-Q2 · question page 1011

LOGIC.INVARIANT.02 · Prove targets impossible

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every light in one whole column. Some sequence of moves leaves exactly one light on.

Answer: False

Reasoning: Look at any 2 by 2 block of lights. A row or column move switches either none or exactly 2 of its 4 lights, so the number of lit lights in the block stays even. With exactly one light on, a block containing it would have 1 lit light, which is odd.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S44-F2-Q3 · question page 1012

LOGIC.INVARIANT.02 · Prove targets impossible

A robot starts at (0, 0). Each move goes from (x, y) to (x + 3, y), to (x, y + 3), or to (x + 1, y + 1). Which point can it reach?

Answer: D. (5, 2)

Reasoning: x - y is always a multiple of 3. (5, 2): two moves of (1, 1) and one of (3, 0).

- ✗ (4, 0): x - y changes only by 3, -3 or 0, so it stays a multiple of 3; here x - y = 4
- ✗ (2, 3): here x - y = -1, not a multiple of 3
- ✗ (1, 5): here x - y = -4, not a multiple of 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S44-F2-Q4 · question page 1013

LOGIC.INVARIANT.02 · Prove targets impossible

A 4 by 4 grid of lights starts all off. A move switches every light in one whole row, or every

light in one whole column. Give a number of lit lights from 1 to 15 that some sequence of moves can leave.

Answer: any values that satisfy: h == 4 or h == 6 or h == 8 or h == 10 or h == 12 (example: h = 6)

Reasoning: Only the rows and columns switched an odd number of times matter. With r such rows and c such columns, the lit count is 4r + 4c - 2rc. For r and c from 0 to 4 this gives 0, 4, 6, 8, 10, 12 or 16.

Tempting answers the rule rejects: h = 1; h = 2; h = 5; h = 14

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Breadth-first search of all reachable patterns: lit counts [0, 4, 6, 8, 10, 12, 16]. Automatic: the example passes the rule and 4 listed mistakes fail it.

S44-F2-Q5 · question page 1014

LOGIC.INVARIANT.02 · Prove targets impossible

The numbers 1, 2, 3, ..., 10 are on a board. A move erases two numbers a and b and writes a + b + 1. The last number left does not depend on which moves are made.

Answer: True

Reasoning: Each move raises the total by exactly 1, and there are always exactly 9 moves, so the last number is 55 + 9 = 64.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-M-Q1 · question page 1016

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Which letter belongs in the shaded cell (bottom row, first column)?

Answer: D

Reasoning: Column 1 already has A (row 1) and C (row 2), so rows 3 and 4 start with B and D in some order. Row 3 already has D (in column 4), so row 3 starts with B and row 4 starts with D. (The whole grid is ADBC, CADB, BCAD, DBCA.)

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-M-Q2 · question page 1017

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is TEN. Give the second and third rows.

Answer: any values that satisfy: w2 == 'era' and w3 == 'nap' (example: w2 = ERA, w3 = NAP)

Reasoning: Column 1 must be TEN, so row 2 starts with E and row 3 with N. Row 2 = column 2, which starts with E: ERA fits (E, R, A). Then column 3 is N, A, ?, and row 3 = NA?: NAP. Check: columns TEN, ERA, NAP.

Tempting answers the rule rejects: w2 = EAR, w3 = NAP; w2 = ERA, w3 = NET

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search of every pair of listed words: 1 word square(s) start with TEN. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-M-Q3 · question page 1018

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: a 3 by 3 grid is filled with the letters A and B so that no two cells that share a side hold the same letter. The center cell holds A. Then all four corner cells hold the same letter.

Answer: True

Reasoning: The center's four side neighbors must be B, and each corner touches two of those B cells, so each corner is A.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-M-Q4 · question page 1019

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B and C exactly once. The top-left cell holds A and the center cell holds A. Which letter must go in the bottom-right cell?

Answer: A. A

Reasoning: Row 3 needs an A, but columns 1 and 2 already have one. So A goes in column 3 of row 3.

- ✗ B: the grid can be completed only with A there
- ✗ C: the grid can be completed only with A there
- ✗ It could be B or C: A is already in rows 1 and 2 and in columns 1 and 2, so A must go in row 3, column 3

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-M-Q5 · question page 1020

LOGIC.GRID.02 · Fill and analyze letter grids under stated rules and a supplied word list

Rule: each row and each column must contain each of the letters A, B and C exactly once. The first row is A B C. How many ways are there to complete the grid?

Answer: 2

Reasoning: The second row must avoid the letter above each cell: B C A or C A B. Each choice forces the third row. So 2 ways.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-B1-Q1 · question page 1021

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. A row reads A _ C. Which letter fills the gap?

Answer: B

Reasoning: Each row holds A, B and C once, and only B is missing.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-B1-Q2 · question page 1022

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. If column 1 holds A and B, its third cell must hold C.

Answer: True

Reasoning: C is the only letter not yet used in that column.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S45-B1-Q3 · question page 1023

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. A cell's row already holds A and D, and its column holds B. Which letter must go in the cell?

Answer: D. C

Reasoning: Only C is used by neither the row nor the column.

- ✗ A: A is already in the row
- ✗ B: B is already in the column
- ✗ D: D is already in the row

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-B1-Q4 · question page 1024

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. A grid can have the row A B C directly above the row A C B.

Answer: False

Reasoning: Both rows put A in column 1, so column 1 would have A twice.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S45-B1-Q5 · question page 1025

LOGIC.GRID.02 · Strategy: fill the most constrained cell first

Rule: each row and each column must contain each of the letters A, B and C exactly once. Rows 1 and 2 are A B C and C A B. What is row 3? Type the three letters with no spaces.

Answer: BCA

Reasoning: Each column needs its missing letter: column 1 has A, C so B; column 2 has B, A so C; column 3 has C, B so A.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q1 · question page 1026

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. The first row of a word square is ATE. Give the second and third rows.

Answer: any values that satisfy: w2 == 'tea' and w3 == 'ear' (example: w2 = TEA, w3 = EAR)

Reasoning: Row 2 = column 2 starts with T, and row 3 = column 3 starts with E. TEA and EAR fit: columns ATE, TEA, EAR.

Tempting answers the rule rejects: w2 = TEN, w3 = EAR; w2 = TAN, w3 = ERA

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: 1 word square(s) start with ATE. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-F1-Q2 · question page 1027

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. How many words on the list can be the first row of at

least one word square?

Answer: 15

Reasoning: A search of every pair of words for each possible first row finds the squares; count the first rows that work.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q3 · question page 1028

LOGIC.GRID.02 · Word squares from a list

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. A word square can start with the row ONE.

Answer: False

Reasoning: Row 2 must start with N (NET, NAP or NOT) and row 3 with E. The middle letter of row 3 must equal the last letter of row 2 (T or P), but no listed word starts with ET or EP.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-F1-Q4 · question page 1029

LOGIC.GRID.02 · Word squares from a list

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. The grid in the figure has six letters filled in. Type the complete top row (four letters, no spaces).

Answer: ADBC

Reasoning: Row 2 is C _ D B, so its gap is A. Column 4 needs A in row

4. Continuing cell by cell gives the only completion: ADBC, CADB, BCAD, DBCA.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F1-Q5 · question page 1030

LOGIC.GRID.02 · Word squares from a list

Rule: a 2 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. How many such grids are there?

Answer: A. 2

Reasoning: The top-left letter decides the whole checkerboard pattern: 2 grids.

- ✗ 1: the top-left cell can be either letter
- ✗ 4: once the top-left cell is chosen, every other cell is forced
- ✗ 6: that counts cells, not grids

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-F2-Q1 · question page 1031

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B, C and D exactly once. How many ways are there to fill the whole grid if the first row is A B C D?

Answer: 24

Reasoning: Count the second rows that avoid the letter above each cell (9 ways), then the completions of each (search): 24 in all.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S45-F2-Q2 · question page 1032

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: a 3 by 3 grid is filled with A and B so that no two cells sharing a side hold the same letter. The top-left cell holds A and the top-right cell holds B.

Answer: False

Reasoning: The top row must alternate A, B, A, so its two ends match. A grid with A and B at the two top corners breaks the rule.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S45-F2-Q3 · question page 1033

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B and C exactly once. Which partly filled grid can NOT be completed?

Answer: B. Row 1 starts A B, and row 2 ends in C

Reasoning: Row 1 is A B _, so its last cell must be C. But column 3 already has C in row 2, so no letter fits.

- ✗ Row 1 is A B C: it can be completed in 2 ways
- ✗ Row 1 starts A, and row 2 starts B: for example A B C, B C A, C A B
- ✗ The diagonal from top left to bottom right is A, A, A: for example A B C, C A B, B C A

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S45-F2-Q4 · question page 1034

LOGIC.GRID.02 · Prove a grid is impossible or count it

Word list (the ONLY words allowed): CAT, ARE, TEN, ART, EAR, NET, TAN, ATE, RAT, ERA, TEA, ACE, ONE, NAP, APE, PEN, OAR, ORE, ROE, TOE, TON, NOT, ANT. In a word square, the three rows, read left to right, are the same words as the three columns, read top to bottom, in the same order. Give the second and third rows of the word square whose first row is ROE.

Answer: any values that satisfy: w2 == 'oar' and w3 == 'era' (example: w2 = OAR, w3 = ERA)

Reasoning: Row 2 starts with O and row 3 with E. OAR and ERA give columns ROE, OAR, ERA.

Tempting answers the rule rejects: w2 = ORE, w3 = ERA; w2 = OAR, w3 = EAR

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: 1 word square(s) start with ROE. Automatic: the example passes the rule and 2 listed mistakes fail it.

S45-F2-Q5 · question page 1035

LOGIC.GRID.02 · Prove a grid is impossible or count it

Rule: each row and each column must contain each of the letters A, B and C exactly once. How many ways are there to fill an empty 3 by 3 grid?

Answer: 12

Reasoning: The first row can be any of the 6 orders of A, B, C, and each has 2 completions: 12.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S46-M-Q1 · question page 1037

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The school's robot passed every test in the lab, so it will work at the competition." Which assumption does this argument need?

Answer: C. Conditions at the competition are like those in the lab.

Reasoning: Lab success supports competition success only if the two settings are alike.

- ✗ The robot was built by students.: who built it does not connect the lab results to the competition
- ✗ The robot failed some tests last year.: that is about the past and weakens nothing here
- ✗ Other teams also tested their robots.: other teams' testing is irrelevant to this robot

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q2 · question page 1038

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Mia's plant grew faster after she moved it to the window, so sunlight made it grow faster." Which assumption does this argument need?

Answer: A. Nothing else that affects growth changed when the plant was moved.

Reasoning: If the temperature or watering also changed, those could explain the faster growth instead.

- ✗ Plants need water to grow.: true, but it does not link the move to the faster growth

✗ The window faces south.: the direction is not needed for the conclusion

✗ Mia has other plants indoors.: other plants are not part of the argument

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q3 · question page 1039

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every member of the chess club is in grade 10. Jo is in grade 10. So Jo is in the chess club." This argument would become valid if we add the assumption "Everyone in grade 10 is in the chess club."

Answer: True

Reasoning: With that added premise, Jo being in grade 10 forces Jo into the club. The original premise points the other way.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-M-Q4 · question page 1040

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

The cafeteria sold fewer sandwiches this week than last week. Which explanation is best supported if we also learn that the cafeteria was closed on two days this week?

Answer: B. The cafeteria was open on fewer days to sell sandwiches.

Reasoning: The given fact directly explains fewer sales without any other change.

✗ Students stopped liking sandwiches.: nothing given says tastes changed

✗ The sandwich price went up.: nothing given mentions prices

✗ Fewer students attend the school this year.: the comparison is week to week, and enrollment was not mentioned

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-M-Q5 · question page 1041

LOGIC.EVIDENCE.01 · Identify the unstated assumption an argument needs and evaluate competing explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "This bridge is safe because it has stood for 80 years." The argument needs the assumption that the bridge will be used the same way as it has been.

Answer: True

Reasoning: If much heavier loads start crossing, 80 years of standing would say little about safety under the new loads.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q1 · question page 1042

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Sam is taller than Ana, so Sam is taller than Raj." Which assumption fills the gap?

Answer: D. Ana is at least as tall as Raj.

Reasoning: Sam > Ana and Ana >= Raj give Sam > Raj.

✗ Raj is taller than Ana.: that would make the conclusion less certain, not more

✗ Sam is the tallest in his class.: Raj may not be in Sam's class
✗ Ana and Raj are friends.: friendship says nothing about height

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-B1-Q2 · question page 1043

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. Every true statement about the topic of an argument is an assumption that the argument needs.

Answer: False

Reasoning: A true fact that does not bridge the gap between premise and conclusion is not needed by the argument.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q3 · question page 1044

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "The park was crowded on Saturday, so it will be crowded this Saturday." What is the gap?

Answer: C. From one past Saturday to a future Saturday

Reasoning: The argument needs this Saturday to be like the last one.

✗ From the park to the city: the argument never mentions a city
✗ From crowded to empty: the conclusion also says crowded
✗ From Saturday to Sunday: both days in the argument are Saturdays

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-B1-Q4 · question page 1045

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. If an assumption an argument needs turns out to be false, the conclusion no longer follows from the premises.

Answer: True

Reasoning: That is what it means for the argument to need it.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-B1-Q5 · question page 1046

LOGIC.EVIDENCE.01 · Find the gap between premise and conclusion

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "All squares are rectangles, so all rectangles are squares" needs no assumption to be valid.

Answer: False

Reasoning: It reverses the premise. It would need the extra premise "all rectangles are squares", which is false.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F1-Q1 · question page 1047

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Claim: "Students who eat breakfast score higher on tests, so eating breakfast raises test scores." Which fact would most WEAKEN the claim?

Answer: D. Students who eat breakfast also sleep more, and sleep is

linked to higher scores.

Reasoning: Another factor (sleep) could explain both, so breakfast might not be the cause.

- ✗ Many students eat breakfast.: how common breakfast is does not affect the link
- ✗ Test scores rose last year.: a general rise does not touch the breakfast link
- ✗ Breakfast foods are cheap.: price does not touch the link between breakfast and scores

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F1-Q2 · question page 1048

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Claim: "The new crosswalk signal reduced accidents at Elm Street." Which fact would most STRENGTHEN the claim?

Answer: C. Accidents did not drop at similar streets that did not get the signal.

Reasoning: If accidents fell only where the signal was added, the signal is the likely cause.

- ✗ The signal cost a lot of money.: cost says nothing about the effect
- ✗ Elm Street is near a school.: location alone does not show the signal worked
- ✗ Drivers like the new signal.: liking it does not show accidents fell because of it

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F1-Q3 · question page 1049

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "No reptiles have fur. This animal has fur. So this animal is not a reptile." This argument needs an unstated assumption to be valid.

Answer: False

Reasoning: It is already valid: if it were a reptile, it would have no fur, but it has fur.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F1-Q4 · question page 1050

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Our team lost every game we played in the rain. It will rain on Friday. So we will lose Friday's game." Which assumption does the argument need?

Answer: B. The rain caused the losses, rather than something else about those games.

Reasoning: The past pattern predicts Friday only if rain is the reason for the losses.

- ✗ Our team has played in the rain before.: that is already stated in the premise
- ✗ Friday's game is at home.: location is not part of the argument
- ✗ It rains more in the spring.: that does not connect rain to losing

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F1-Q5 · question page 1051

LOGIC.EVIDENCE.01 · Weaken and strengthen explanations

Two explanations both fit the facts. Then no new fact can help decide between them.

Answer: False

Reasoning: A new fact that one explanation predicts and the other does not supports the first over the second.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F2-Q1 · question page 1052

LOGIC.EVIDENCE.01 · Judge explanations with care

A town's ice-cream sales and its number of sunburns rise and fall together during the year. Which explanation is best?

Answer: A. Hot, sunny weather increases both.

Reasoning: A common cause (weather) explains why the two move together.

- ✗ Eating ice cream causes sunburn.: nothing suggests ice cream affects skin
- ✗ Sunburns make people buy ice cream.: the timing fits weather better than this
- ✗ The two numbers match by chance every year.: a yearly pattern is unlikely to be chance alone

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S46-F2-Q2 · question page 1053

LOGIC.EVIDENCE.01 · Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Ming read the book, so Ming knows how it ends." Which assumption is needed?

Answer: B. Ming read the book all the way to the end.

Reasoning: Reading part of the book would not reveal the ending.

- ✗ The book has a surprise ending.: the type of ending does not matter
- ✗ Ming liked the book.: liking it is not needed to know the ending
- ✗ The book is long.: length is not needed for the conclusion

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F2-Q3 · question page 1054

LOGIC.EVIDENCE.01 · Judge explanations with care

An assumption is an unstated statement that an argument needs in order for its conclusion to follow. "Every prize winner practiced daily. Kai practiced daily. So Kai won a prize." Adding "Only prize winners practiced daily" makes the argument valid.

Answer: True

Reasoning: Then anyone who practiced daily is a prize winner, and Kai practiced daily.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S46-F2-Q4 · question page 1055

LOGIC.EVIDENCE.01 · Judge explanations with care

A survey of the chess club finds that 90% of members prefer quiet study rooms. A student concludes that 90% of the whole school prefers quiet study rooms. What assumption does this need?

Answer: C. Chess club members have the same preferences as the whole school.

Reasoning: The conclusion jumps from one group to the school; it needs the group to represent the school.

- ✗ The chess club has more than 10 members.: size alone does not make the club like the school
- ✗ Quiet rooms are better for studying.: that does not link the club to the school
- ✗ The survey asked about study rooms.: that is already part of the information

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S46-F2-Q5 · question page 1056

LOGIC.EVIDENCE.01 · Judge explanations with care

If a needed assumption is shown to be false, the conclusion of the argument must be false.

Answer: False

Reasoning: The argument fails, but the conclusion could still be true for other reasons.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S47-M-Q1 · question page 1058

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Ana, Ben and Cal ran a race with no ties. Who finished first? Clue (1): Ana finished ahead of Ben. Clue (2): Cal finished last.

Answer: A. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows A B C, A C B and C A B (first could be Ana or Cal). Clue (2) allows A B C and B A C. Together only A B C: Ana.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q2 · question page 1059

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) x is a whole number from 1 to 20. What is x? Clue (1): x is a multiple of 7 and greater than 10. Clue (2): x is even.

Answer: D. Clue (1) alone is enough, but clue (2) alone is not.

Reasoning: Clue (1) leaves only 14. Clue (2) leaves 10 numbers.

- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q3 · question page 1060

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Two whole numbers, each from 1 to 9, add up to 17. Knowing only this is enough to know which two numbers they are.

Answer: True

Reasoning: Only 8 + 9 = 17 (in either order).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-M-Q4 · question page 1061

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) A rectangle has whole-number side lengths. What is its area? Clue (1): its perimeter is 12. Clue (2): one side is twice as long as the other.

Answer: B. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows 1 by 5, 2 by 4, 3 by 3 (areas 5, 8, 9). Clue (2) allows 1 by 2, 2 by 4, and more. Together: 2 by 4, area 8.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-M-Q5 · question page 1062

LOGIC.EVIDENCE.02 · Decide whether the given information determines the answer, and what extra fact would

x is a whole number from 1 to 50. You know that x is a multiple of 6 and also a multiple of 4. How many values of x are still possible?

Answer: 4

Reasoning: Multiples of both 6 and 4 are multiples of 12: 12, 24, 36, 48. Four values remain, so the clues are not enough to find x.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-B1-Q1 · question page 1063

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 10, and n is a multiple of 4. How many values of n are possible?

Answer: 2

Reasoning: 4 and 8.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-B1-Q2 · question page 1064

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 10. Knowing that n is prime and even is enough to find n.

Answer: True

Reasoning: 2 is the only even prime.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-B1-Q3 · question page 1065

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

Two whole numbers add to 10. Knowing only this is enough to find the larger number.

Answer: False

Reasoning: 1 + 9, 2 + 8, 3 + 7, 4 + 6 and 5 + 5 all work.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S47-B1-Q4 · question page 1066

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

A clue is "enough" to answer a question when...

Answer: A. exactly one answer is left after using it.

Reasoning: Enough means the answer is determined.

- ✗ at least one answer is left after using it.: two or more answers left means not enough
- ✗ it rules out at least one answer.: it may still leave several answers
- ✗ it is true.: a true clue can still leave several answers

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S47-B1-Q5 · question page 1067

LOGIC.EVIDENCE.02 · Strategy: list what each clue leaves

n is a whole number from 1 to 20. Clue: n is a square number and n is odd. How many values of n are left?

Answer: 2

Reasoning: 1 and 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S47-F1-Q1 · question page 1068

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Four friends sit in a row of four seats numbered 1 to 4: Dee, Eli, Fay and Gus. Who sits in seat 1? Clue (1): Dee sits in seat 4. Clue (2): Eli sits directly left of Fay, and Gus sits directly left of Eli.

Answer: C. Neither clue alone is enough, but both together are.

Reasoning: Clue (2) forces G E F together in that order, so the order is G E F D or D G E F: not enough. Clue (1) alone leaves 6 orders. Together: G E F D, so Gus.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q2 · question page 1069

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) Pat is older than Quinn. How old is Quinn, in whole years? Clue (1): their ages add to 30. Clue (2): Pat is 4 years older than Quinn.

Answer: D. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows many pairs (16, 14), (20, 10), ... Clue (2) allows many pairs. Together: P + Q = 30 and P - Q = 4, so Q = 13.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q3 · question page 1070

LOGIC.EVIDENCE.02 · Which clue is enough?

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 30. What is n? Clue (1): n leaves remainder 1 when divided by 7. Clue (2): n is a perfect square.

Answer: C. Neither clue alone is enough, but both together are.

Reasoning: Clue (1) allows 1, 8, 15, 22, 29. Clue (2) allows 1, 4, 9, 16,

25. Only 1 is in both lists.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F1-Q4 · question page 1071

LOGIC.EVIDENCE.02 · Which clue is enough?

A bag holds only red and blue marbles, 12 in all. Knowing that there are 4 more red marbles than blue ones is enough to find the number of red marbles.

Answer: True

Reasoning: r + b = 12 and r - b = 4 give r = 8.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-F1-Q5 · question page 1072

LOGIC.EVIDENCE.02 · Which clue is enough?

x is a whole number from 1 to 20 and x is a multiple of 3. Give one extra clue as a single number: a value m from 2 to 10 such that knowing "x is also a multiple of m" leaves exactly one possible x.

Answer: any values that satisfy: m == 4 or m == 5 (example: m = 4)

Reasoning: The multiples of 3 up to 20 are 3, 6, 9, 12, 15, 18. "Also a multiple of 4" leaves only 12, and "also a multiple of 5" leaves only 15.

Other values of m leave none or several.

Tempting answers the rule rejects: m = 2; m = 6; m = 9; m = 7

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every m from 2 to 10: exactly one x remains for m = [4, 5]. Automatic: the example passes the rule and 4 listed mistakes fail it.

S47-F2-Q1 · question page 1073

LOGIC.EVIDENCE.02 · Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n + 5 = 12. Clue (2): 2n = 14.

Answer: D. Each clue alone is enough.

Reasoning: Each clue alone gives n = 7.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Clue (2) alone is enough, but clue (1) alone is not.: check whether clue (2) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F2-Q2 · question page 1074

LOGIC.EVIDENCE.02 · Design the missing clue

Decide whether each clue, alone or together with the other, is enough to answer the question with exactly one answer. (In each of these questions, the two clues together are enough.) n is a whole number from 1 to 20. What is n? Clue (1): n is odd. Clue (2): n is a

multiple of 13.

Answer: A. Clue (2) alone is enough, but clue (1) alone is not.

Reasoning: Clue (2) alone leaves only 13 (26 is too big). Clue (1) leaves 10 odd numbers.

- ✗ Clue (1) alone is enough, but clue (2) alone is not.: check whether clue (1) alone leaves exactly one answer
- ✗ Neither clue alone is enough, but both together are.: check each clue alone before combining them
- ✗ Each clue alone is enough.: one of the clues alone leaves more than one answer

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S47-F2-Q3 · question page 1075

LOGIC.EVIDENCE.02 · Design the missing clue

Two whole numbers from 1 to 9 multiply to 12. Give a sum s such that also knowing "the two numbers add to s" tells you exactly which two numbers they are.

Answer: any values that satisfy: s == 7 or s == 8 (example: s = 7)

Reasoning: The pairs with product 12 are 2 and 6 (sum 8) and 3 and 4 (sum 7). Either sum picks out one pair.

Tempting answers the rule rejects: s = 13; s = 6; s = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every sum from 2 to 18: the pair is determined for s = [7, 8]. Automatic: the example passes the rule and 3 listed mistakes fail it.

S47-F2-Q4 · question page 1076

LOGIC.EVIDENCE.02 · Design the missing clue

Two whole numbers, each from 1 to 9, multiply to 36. Knowing only this is enough to know which two numbers they are.

Answer: False

Reasoning: Both 4 x 9 and 6 x 6 equal 36.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S47-F2-Q5 · question page 1077

LOGIC.EVIDENCE.02 · Design the missing clue

Three different whole numbers from 1 to 9 add up to 23. How many different sets of three numbers are possible?

Answer: 1

Reasoning: The largest possible total is 9 + 8 + 7 = 24. To get 23, lower one number by 1 without repeating: only 9 + 8 + 6. So the total alone is enough to find the set.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-M-Q1 · question page 1079

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments are lit when the display shows 2026?

Answer: 22

Reasoning: 2 uses 5, 0 uses 6, 2 uses 5, 6 uses 6: 22.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-M-Q2 · question page 1080

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit uses exactly 3 lit segments?

Answer: B. 7

Reasoning: 7 lights the top bar and the two right-hand segments.

- ✗ 1: 1 uses 2 segments
- ✗ 4: 4 uses 4 segments
- ✗ 2: 2 uses 5 segments

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-M-Q3 · question page 1081

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. When the display showing 69 is turned upside down, it shows 69 again.

Answer: True

Reasoning: A half turn turns 6 into 9 and 9 into 6, and also reverses their order: 69 becomes 69.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-M-Q4 · question page 1082

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit that looks exactly the same after the display is turned upside down.

Answer: any values that satisfy: d == 0 or d == 2 or d == 5 or d == 8 (example: d = 8)

Reasoning: 0, 2, 5 and 8 are unchanged by a half turn. 6 and 9 swap, and 1, 3, 4, 7 do not become digits.

Tempting answers the rule rejects: d = 6; d = 1; d = 9; d = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked by turning every digit: unchanged digits ['0', '2', '5', '8']. Automatic: the example passes the rule and 4 listed mistakes fail it.

S48-M-Q5 · question page 1083

LOGIC.SPATIAL.06 · Reason about seven-segment digits: segments used, rotations and segment moves

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the digit 2 is moved to an unlit position. Which digit can result?

Answer: D. 3

Reasoning: Move the lower-left segment to the lower-right position: 2 becomes 3.

- ✗ 5: turning 2 into 5 needs two segments moved
- ✗ 8: 8 has more segments than 2
- ✗ 7: 7 has fewer segments than 2

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-B1-Q1 · question page 1084

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many segments does the digit 4 light?

Answer: 4

Reasoning: Upper left, middle, and both right-hand segments: 4.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-B1-Q2 · question page 1085

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 8 lights all seven segments.

Answer: True

Reasoning: 8 uses every segment.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-B1-Q3 · question page 1086

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Which digit lights the fewest segments?

Answer: C. 1

Reasoning: 1 lights only 2 segments.

- ✗ 7: 7 lights 3
- ✗ 4: 4 lights 4
- ✗ 0: 0 lights 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-B1-Q4 · question page 1087

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digits 0 and 6 light the same number of segments.

Answer: True

Reasoning: Each lights 6 segments (in different positions).

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-B1-Q5 · question page 1088

LOGIC.SPATIAL.06 · Read the segments of each digit

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The digit 3 lights the middle segment and both left-hand segments.

Answer: False

Reasoning: 3 lights the top, middle, bottom and both right-hand segments; no left-hand ones.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F1-Q1 · question page 1089

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display shows 1958. How many segments are lit?

Answer: 20

Reasoning: 2 + 6 + 5 + 7 = 20.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F1-Q2 · question page 1090

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. The display shows 625. What does it show after being turned upside down?

Answer: A. 529

Reasoning: Reverse the order and turn each digit: 5 stays 5, 2 stays 2, 6 becomes 9: 529.

- ✗ 526: the 6 also turns into a 9
- ✗ 925: the order of the digits is also reversed
- ✗ It does not show a number: 6, 2 and 5 all turn into digits

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-F1-Q3 · question page 1091

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. The display showing 1001 still shows a number after it is turned upside down.

Answer: False

Reasoning: The 1 lights the two right-hand segments; turned, they become the two left-hand segments, which is not a digit in the reference.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F1-Q4 · question page 1092

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. How many two-digit numbers

(10 to 99) still show a number after the display is turned upside down?

Answer: 30

Reasoning: Both digits must be from 0, 2, 5, 6, 8, 9; the first cannot be 0: 5 x 6 = 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F1-Q5 · question page 1093

LOGIC.SPATIAL.06 · Turned displays and segment counts

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Give a digit d such that adding exactly one lit segment to d (moving nothing) makes the digit 8.

Answer: any values that satisfy: d == 0 or d == 6 or d == 9 (example: d = 0)

Reasoning: 0, 6 and 9 each light 6 segments, and each is 8 with one segment missing.

Tempting answers the rule rejects: d = 5; d = 3; d = 8

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: digits one segment short of 8: ['0', '6', '9']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S48-F2-Q1 · question page 1094

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. Exactly one lit segment of the

digit 3 is moved to an unlit position. Give a digit that can result.

Answer: any values that satisfy: d == 2 or d == 5 (example: d = 5)

Reasoning: Move the upper-right segment to the upper left to make 5, or the lower-right segment to the lower left to make 2.

Tempting answers the rule rejects: d = 9; d = 8; d = 3

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every one-segment move: 3 can become ['2', '5']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S48-F2-Q2 · question page 1095

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. You may light at most 12 segments in total. What is the largest whole number you can show? (Any number of digits; the first digit is not 0.)

Answer: 111111

Reasoning: More digits beat bigger digits, and 1 is the cheapest digit (2 segments). Six 1s use 12 segments: 111111. Swapping a 1 for 7 costs one more segment, which is not available.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S48-F2-Q3 · question page 1096

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. Every digit can be turned into some other digit by moving exactly one lit segment.

Answer: False

Reasoning: 1, 4, 7 and 8 cannot: no single move turns them into another digit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S48-F2-Q4 · question page 1097

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its lit segments are exactly those of a digit in the reference figure. A clock shows hours 1 to 12 and minutes 00 to 59 (for example 9:58 uses the digits 9, 5, 8). Which time lights the most segments?

Answer: B. 10:08

Reasoning: 10:08 lights 2 + 6 + 6 + 7 = 21. A search of every time confirms 21 is the most, and 10:08 is one of the times that reach it.

- ✗ 12:59: 12:59 lights 2 + 5 + 5 + 6 = 18
- ✗ 9:58: 9:58 lights 6 + 5 + 7 = 18
- ✗ 11:11: 11:11 lights only 8

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S48-F2-Q5 · question page 1098

LOGIC.SPATIAL.06 · Segment moves and best displays

Digits are shown on a seven-segment display exactly as in the reference figure (0 to 9). Turning the display upside down means a half turn; a turned digit counts as a digit only if its

lit segments are exactly those of a digit in the reference figure. How many digits light more than 5 segments?

Answer: 4

Reasoning: 0, 6, 9 light 6 and 8 lights 7: four digits.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S49-M-Q1 · question page 1100

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Hasty generalization: drawing a broad conclusion from too few cases. Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Dana argues that the library should stay open late

Answer: D. Attacking the person

Reasoning: Raj rejects Dana's claim because of something about Dana, not because of her reasons.

- ✗ Hasty generalization: Raj does not generalize from a few cases
- ✗ Circular reasoning: Raj does not use his conclusion as a reason
- ✗ Appeal to popularity: Raj does not mention what many people believe

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-M-Q2 · question page 1101

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Straw man: attacking a distorted version of someone's argument instead of the real one. After, so

because: concluding that A caused B only because B happened after A. "Either we cancel the field trip, or every student wil

Answer: A. False choice

Reasoning: Other options exist, such as more chaperones or a planned route.

- ✗ Slippery slope: no chain of steps is described
- ✗ Straw man: no one's argument is being distorted
- ✗ After, so because: no claim about one event following another is made

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-M-Q3 · question page 1102

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Circular reasoning: using the conclusion, in other words, as a reason for itself. "I met two rude people from that town, so everyone from that to

Answer: C. Hasty generalization

Reasoning: Two cases are far too few to support a claim about everyone in a town.

- ✗ Attacking the person: no one's argument is being rejected
- ✗ False choice: no two options are presented
- ✗ Circular reasoning: the conclusion is not used as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S49-M-Q4 · question page 1103

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. "This app is the best because it is better than all the others" is an example of circular reasoning.

Answer: True

Reasoning: "Better than all the others" says the same thing as "the best", so the reason is just the conclusion restated.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-M-Q5 · question page 1104

LOGIC.FALLACY.01 · Recognize common informal fallacies in self-contained arguments

Definitions: After, so because: concluding that A caused B only because B happened after A. Hasty generalization: drawing a broad conclusion from too few cases. Appeal to popularity: claiming something is true because many people believe it. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "After I started wearing my lucky socks, our team won three

Answer: B. After, so because

Reasoning: The wins came after the socks, but nothing shows the socks caused them.

- ✗ Hasty generalization: it does not generalize to a group
- ✗ Appeal to popularity: it does not mention what many people believe
- ✗ Slippery slope: it does not predict a chain of extreme results

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q1 · question page 1105

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. False choice: presenting only two options when others are possible. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Which definition describes "Millions of people watch

Answer: B. Appeal to popularity

Reasoning: The only reason given is how many people watch it.

- ✗ Straw man: no one's argument is distorted
- ✗ False choice: no two options are presented
- ✗ Slippery slope: no chain of results is predicted

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q2 · question page 1106

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. Concluding that all dogs are friendly after meeting three friendly dogs is a hasty generalization.

Answer: True

Reasoning: Three dogs are too few cases for a claim about all dogs.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-B1-Q3 · question page 1107

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Saying "The bridge design is unsafe because the beams are

too thin for the load" is attacking the person.

Answer: False

Reasoning: The reason is about the design itself, not about a person.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-B1-Q4 · question page 1108

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

An argument's reason is about who made a claim rather than about the claim. Which question best helps you notice this?

Answer: A. Is the reason about the claim, or about something else?

Reasoning: Checking what the reason is about exposes attacks on the person.

- ✗ Is the conclusion surprising?: surprising conclusions can be well supported
- ✗ Is the argument long?: length does not reveal what the reasons are about
- ✗ Do I agree with the conclusion?: agreeing does not show whether the reasons are good

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-B1-Q5 · question page 1109

LOGIC.FALLACY.01 · Strategy: name what the argument does instead of giving reasons

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. "If we allow phones at lunch, then students will use them in class, then stop learning, then fail school" is a slippery slope.

Answer: True

Reasoning: A chain of extreme results is claimed without support for each step.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F1-Q1 · question page 1110

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. False choice: presenting only two options when others are possible. Hasty generalization: drawing a broad conclusion from too few cases. Ava says the school should add one vegetarian meal option

Answer: C. Straw man

Reasoning: Ben attacks a much stronger claim (a full ban) than the one Ava made.

- ✗ Attacking the person: Ben attacks a claim, not Ava's character
- ✗ False choice: Ben does not offer only two options
- ✗ Hasty generalization: Ben does not generalize from a few cases

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q2 · question page 1111

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. After, so because: concluding that A caused B only because B happened after A. Circular reasoning: using the conclusion, in other words, as a reason for itself. "If we let one student retake the quiz, soon everyone will r

Answer: D. Slippery slope

Reasoning: It predicts an extreme chain of results from one step, without support.

- ✗ False choice: it does not present exactly two options
- ✗ After, so because: it does not claim a past event caused another
- ✗ Circular reasoning: the conclusion is not used as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q3 · question page 1112

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Appeal to popularity: claiming something is true because many people believe it. "Most dentists in a large survey recommend brushing twice a day, so brushing twice a day is likely a good idea" is an appeal to popularity.

Answer: False

Reasoning: The people cited are experts on teeth, and their recommendation is evidence about the topic. An appeal to popularity relies only on how many people believe something.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F1-Q4 · question page 1113

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: Circular reasoning: using the conclusion, in other words, as a reason for itself. Appeal to popularity: claiming something is true because many people believe it. Straw man: attacking a distorted version of someone's argument instead of the real one. Hasty generalization: drawing a broad conclusion from too few cases. "The rule must be fair because it is the rule, and rules are fair."

Answer: C. Circular reasoning

Reasoning: Being a rule is offered as proof of fairness only because rules are assumed to be fair: the conclusion supports itself.

- ✗ Appeal to popularity: it does not mention what many people believe
- ✗ Straw man: it does not distort anyone's argument
- ✗ Hasty generalization: it does not reason from a few examples

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F1-Q5 · question page 1114

LOGIC.FALLACY.01 · Straw men, slopes and popularity

Definitions: After, so because: concluding that A caused B only because B happened after A. Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. False choice: presenting only two options when others are possible. "The new principal started in Septemb

Answer: B. After, so because

Reasoning: The timing alone is used as proof that one caused the other.

- ✗ Attacking the person: no argument by the principal is being rejected
- ✗ Slippery slope: no chain of future results is predicted
- ✗ False choice: no two options are presented

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q1 · question page 1115

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Attacking the person: rejecting a claim because of who made it, instead of because of the reasons given. Which reply is NOT attacking the person?

Answer: A. "Your numbers are from 2010, and the prices have changed since then."

Reasoning: The correct reply challenges the evidence itself.

- ✗ "You are too young to understand prices.": it dismisses the claim because of the speaker's age
- ✗ "You only say that because you work for the store.": it dismisses the claim because of the speaker's job, not the reasons
- ✗ "Nobody listens to you anyway.": it is about the speaker, not the claim

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q2 · question page 1116

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: False choice: presenting only two options when others are possible. "A whole number is either even or odd" is a false choice.

Answer: False

Reasoning: Every whole number really is even or odd, so the two options are complete.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S49-F2-Q3 · question page 1117

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Hasty generalization: drawing a broad conclusion from too few cases. After, so because: concluding that A caused B only because B happened after A. Straw man: attacking a distorted version of someone's argument instead of the real one. Appeal to popularity: claiming something is true because many people believe it. Which argument avoids all four fallacies defined above?

Answer: D. A random sample of 500 voters favored the plan, so most voters probably favor it.

Reasoning: A large random sample is good evidence about the whole group.

- ✗ Our team won after the coach changed shoes, so the shoes helped.: after, so because
- ✗ Everyone online says the plan is good, so it is good.: appeal to popularity
- ✗ My two cousins liked the plan, so most voters like it.: hasty generalization from two people

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S49-F2-Q4 · question page 1118

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Straw man: attacking a distorted version of someone's argument instead of the real one. Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. Appeal to popularity: claiming something is true because many people believe it. Circular reasoning: using the conclusion, in other words, as a reason for itself. Mo says homework should be shorter on

Answer: B. Straw man

Reasoning: Lee replaces Mo's modest claim with an extreme one and attacks that.

- ✗ Slippery slope: Lee does not predict a chain of steps from Mo's idea
- ✗ Appeal to popularity: Lee does not mention what many people think
- ✗ Circular reasoning: Lee does not use the conclusion as a reason

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S49-F2-Q5 · question page 1119

LOGIC.FALLACY.01 · Is it really a fallacy?

Definitions: Slippery slope: claiming, without support, that one step will lead to a chain of extreme results. An argument that predicts a chain of results is always a slippery slope, even when each step is supported by evidence.

Answer: False

Reasoning: A slippery slope is a fallacy only when the steps are claimed without support. A well-supported chain of causes can be good reasoning.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-M-Q1 · question page 1121

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

To learn how often people in a town exercise, a student surveys 100 people leaving a gym. What is the main problem with this evidence?

Answer: A. People leaving a gym probably exercise more than the town as a whole.

Reasoning: The sample is not representative of the town.

- ✗ 100 people is always too few for a survey.: size is not the main issue; who was asked is
- ✗ The survey should have asked about diet too.: that changes the question, not the evidence
- ✗ Surveys can never give useful evidence.: well-chosen samples can be very useful

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

S50-M-Q2 · question page 1122

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

School A: 45 of its 60 students passed a test. School B: 52 of its 80 students passed. What percent of School A's students passed?

Answer: 75

Reasoning: 45 / 60 = 0.75 = 75%. (School B: 52 / 80 = 65%.) Comparing percents, not counts, is fair when the schools differ in size.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-M-Q3 · question page 1123

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

In many towns, more ice-cream sales come with more swimming accidents. This shows that eating ice cream causes swimming accidents.

Answer: False

Reasoning: Both rise in hot weather. A correlation alone does not show that one causes the other.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-M-Q4 · question page 1124

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

Which study design gives the best evidence that a tutoring program raises test scores?

Answer: C. Randomly assign students to get tutoring or not, then compare the two groups' scores.

Reasoning: Random assignment makes the groups alike except for the tutoring.

- ✗ Ask tutored students whether they feel they improved.: feelings are not test scores, and there is no comparison group
- ✗ Compare tutored students with students who chose not to sign up.: students who sign up may differ in other ways
- ✗ Measure the scores of tutored students only.: with no comparison group, the effect is unknown

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-M-Q5 · question page 1125

LOGIC.EVIDENCE.03 · Judge whether evidence supports a claim: samples, comparisons and correlation

A claim says that more than half of a school's students prefer a later start time. In a random sample of 40 students, n said they prefer it. Give a value of n that would support the claim.

Answer: any values that satisfy: isInteger(n) and n > 20 and n <= 40 (example: n = 28)

Reasoning: More than half of 40 is more than 20, so n from 21 to 40 supports the claim.

Tempting answers the rule rejects: n = 20; n = 12; n = 41

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 20 valid answers.

S50-B1-Q1 · question page 1126

LOGIC.EVIDENCE.03 · Compare rates, not counts

18 of 24 students in a class walk to school. What percent is that?

Answer: 75

Reasoning: 18 / 24 = 0.75 = 75%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-B1-Q2 · question page 1127

LOGIC.EVIDENCE.03 · Compare rates, not counts

Club X has 30 members and 12 play chess. Club Y has 10 members and 6 play chess. A larger share of Club Y plays chess.

Answer: True

Reasoning: X: 12 / 30 = 40%. Y: 6 / 10 = 60%.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S50-B1-Q3 · question page 1128

LOGIC.EVIDENCE.03 · Compare rates, not counts

Town P had 50 car accidents and Town Q had 30. What else do you need before saying Town P's roads are more dangerous?

Answer: D. How many people or cars each town has

Reasoning: Accidents per car or per person is the fair comparison.

- ✗ The names of the drivers: names do not affect the rate
- ✗ Nothing; 50 is more than 30: a larger town can have more accidents and still be safer

✗ The colors of the cars: color is not relevant to the comparison

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-B1-Q4 · question page 1129

LOGIC.EVIDENCE.03 · Compare rates, not counts

A survey of 200 people finds that 30 own a bike. What percent own a bike?

Answer: 15

Reasoning: 30 / 200 = 15%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-B1-Q5 · question page 1130

LOGIC.EVIDENCE.03 · Compare rates, not counts

If Group A has more members with a trait than Group B, then a larger percent of Group A has the trait.

Answer: False

Reasoning: Group A might be much larger: 30 of 100 is less than 20 of 40.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F1-Q1 · question page 1131

LOGIC.EVIDENCE.03 · Samples and comparisons

A website asks visitors to vote on whether the town needs a new park. Of 300 votes, 80% say yes. Why is this weak evidence about the whole town?

Answer: D. People who choose to vote online may care more about parks than others.

Reasoning: A self-selected sample can differ from the population.

- ✗ 300 votes is more than enough.: that does not address who chose to vote
- ✗ 80% is too high to be believed.: a high percent can be accurate
- ✗ Websites cannot count votes.: counting is not the problem

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-F1-Q2 · question page 1132

LOGIC.EVIDENCE.03 · Samples and comparisons

Class A: 21 of 28 students finished the project on time. Class B: 24 of 30. What is the higher of the two on-time percents?

Answer: 80

Reasoning: A: 75%. B: 80%. The higher is 80%, even though the counts are close.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F1-Q3 · question page 1133

LOGIC.EVIDENCE.03 · Samples and comparisons

A study finds that students who own more books tend to have higher reading scores. This alone proves that buying books raises reading scores.

Answer: False

Reasoning: Families who buy many books may also read together more.

The correlation alone does not prove cause.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F1-Q4 · question page 1134

LOGIC.EVIDENCE.03 · Samples and comparisons

To test whether a new fertilizer helps tomato plants grow, which design is best?

Answer: A. Give fertilizer to a randomly chosen half of the plants and compare them with the other half.

Reasoning: Random assignment and a comparison group isolate the effect of the fertilizer.

- ✗ Give fertilizer to all plants and measure them.: there is no comparison group
- ✗ Use the fertilizer only on the healthiest plants.: healthier plants would grow more anyway
- ✗ Ask a gardener whether the plants look bigger.: opinion is not a measurement

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S50-F1-Q5 · question page 1135

LOGIC.EVIDENCE.03 · Samples and comparisons

A coin is flipped 50 times to test whether it is fair. Give a number of heads h from 0 to 50 that would be strong evidence the coin favors heads. Use this test: h is at least 35.

Answer: any values that satisfy: isInteger(h) and h >= 35 and h <= 50 (example: h = 40)

Reasoning: The stated test counts 35 or more heads as strong evidence.

(A fair coin rarely gives that many.)

Tempting answers the rule rejects: h = 26; h = 30; h = 51

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 16 valid answers.

S50-F2-Q1 · question page 1136

LOGIC.EVIDENCE.03 · Comparisons that reverse

Two clinics treat a condition. Clinic A treats 100 mild cases (90 recover) and 20 severe cases (8 recover). Clinic B treats 20 mild cases (19 recover) and 100 severe cases (50 recover). What percent of all Clinic A patients recover, to the nearest whole percent?

Answer: 82

Reasoning: (90 + 8) / 120 = 98 / 120, about 81.7%, which rounds to 82%.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F2-Q2 · question page 1137

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the clinics above, Clinic A has the higher overall recovery rate (about 82% to 58%). Which statement is true?

Answer: B. Clinic B has the higher recovery rate for mild cases AND for severe cases.

Reasoning: A treats mostly mild cases, which recover more often, so its combined rate is higher even though B does better within each group.

- ✗ Clinic A is better for both mild and severe cases.: B: 95% mild, 50% severe; A: 90% mild, 40% severe

✗ The two clinics are equally good for mild cases.: 90% and 95% are not equal

✗ Clinic A must be the better clinic.: A looks better only because it treats more mild cases

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S50-F2-Q3 · question page 1138

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the clinics above, comparing only the overall recovery rates gives a fair comparison of the clinics.

Answer: False

Reasoning: The clinics treat very different mixes of cases. Comparing within each group is fairer.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S50-F2-Q4 · question page 1139

LOGIC.EVIDENCE.03 · Comparisons that reverse

A test for a rare condition is positive for 95 of every 100 people who have it and for 5 of every 100 people who do not. In a town of 10,000 people, 100 have the condition. About how many people in the town test positive in all?

Answer: 590

Reasoning: Have it: 95 of 100 test positive. Do not have it: 5% of 9,900 = 495. Total 590.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S50-F2-Q5 · question page 1140

LOGIC.EVIDENCE.03 · Comparisons that reverse

In the town above, 590 people test positive and 95 of them have the condition. What fraction of the positive tests are correct?

Answer: C. About 16%

Reasoning: 95 / 590 is about 0.161, or 16%. Most positives come from the much larger group without the condition.

- ✗ About 95%: 95% is the rate among people who have the condition, not among positive tests
- ✗ About 5%: 5% is the false-positive rate among people without it
- ✗ About 50%: 95 is far less than half of 590

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-M-Q1 · question page 1142

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Two whole numbers add up to 20. What is the largest possible product?

Answer: 100

Reasoning: 10 x 10 = 100. Moving one number up by k and the other down by k gives (10 + k)(10 - k) = 100 - k squared, which is smaller.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-M-Q2 · question page 1143

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

A rectangular pen uses exactly 24 meters of fence, and its sides are whole numbers of meters. What is the largest possible area, in square meters?

Answer: 36

Reasoning: The sides add to 12. Listing 1 x 11, 2 x 10, ..., 6 x 6 gives areas 11, 20, 27, 32, 35, 36. The 6 by 6 square is largest.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-M-Q3 · question page 1144

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

A 12-hour digital clock shows times from 1:00 to 12:59. Which time has the largest product of its digits?

Answer: B. 9:59

Reasoning: 9 x 5 x 9 = 405. The hour digit is at most 9 and the minute digits at most 5 and 9, so nothing can beat 405, and only 9:59 reaches it.

- ✗ 12:59: 1 x 2 x 5 x 9 = 90
- ✗ 8:59: 8 x 5 x 9 = 360
- ✗ 11:59: 1 x 1 x 5 x 9 = 45

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-M-Q4 · question page 1145

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Using coins worth 1, 5, 10 and 25 cents, the fewest coins that make exactly 30 cents is 2.

Answer: True

Reasoning: 25 + 5 uses 2 coins, and no single coin is worth 30 cents.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-M-Q5 · question page 1146

LOGIC.OPTIMIZE.01 · Find the best value under stated constraints and show nothing better is possible

Whole numbers x and y are each 0 or more, with x + 2y at most 10 and x at most 6. Make x + 3y as large as possible. Give x and y.

Answer: any values that satisfy: x == 0 and y == 5 (example: x = 0, y = 5)

Reasoning: x + 3y = (x + 2y) + y is at most 10 + y, and y is at most 5, so x + 3y is at most 15. Only x = 0, y = 5 reaches 15.

Tempting answers the rule rejects: x = 6, y = 2; x = 2, y = 4; x = 1, y = 5

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all 30 allowed pairs: the best value is 15, reached only at (0, 5). Automatic: the example passes the rule and 3 listed mistakes fail it. Search of the stated domain: 1 valid answers.

S51-B1-Q1 · question page 1147

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Two whole numbers add to 14. What is the largest possible product?

Answer: 49

Reasoning: 7 x 7 = 49.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-B1-Q2 · question page 1148

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

To show that a value is the largest possible, it is enough to find one example that reaches it.

Answer: False

Reasoning: You also need a reason no other answer can be larger.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S51-B1-Q3 · question page 1149

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Which numbers from the list 4, 9, 2, 7, 5 give the largest sum if you may choose any two?

Answer: C. 9 and 7

Reasoning: The two largest numbers give the largest sum: 16.

- ✗ 9 and 5: 9 + 5 = 14, less than 16
- ✗ 4 and 9: 4 + 9 = 13
- ✗ 7 and 5: 7 + 5 = 12

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S51-B1-Q4 · question page 1150

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

Using coins worth 1, 5 and 10 cents, what is the fewest coins that make 27 cents?

Answer: 5

Reasoning: 10 + 10 + 5 + 1 + 1: 5 coins.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-B1-Q5 · question page 1151

LOGIC.OPTIMIZE.01 · Strategy: find a good answer, then prove nothing beats it

A rectangle with whole-number sides and perimeter 20 can have area 25.

Answer: True

Reasoning: A 5 by 5 square.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F1-Q1 · question page 1152

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Coins are worth 1, 4 and 6 cents. What is the fewest coins that make exactly 8 cents?

Answer: 2

Reasoning: Taking the biggest coin first gives 6 + 1 + 1 (3 coins), but 4 + 4 uses only 2.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F1-Q2 · question page 1153

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Whole numbers add up to 10 (you choose how many numbers). What is the largest possible product?

Answer: 36

Reasoning: Use 3s and 2s: 3 + 3 + 2 + 2 gives 36, and 3 + 3 + 4 also gives 36. Splitting any part 5 or more into smaller parts raises the product, and 1s never help.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F1-Q3 · question page 1154

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Coins are worth 1, 4 and 6 cents. Always taking the largest coin that fits gives the fewest coins for every amount.

Answer: False

Reasoning: For 8 cents it gives 6 + 1 + 1, but 4 + 4 is better.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F1-Q4 · question page 1155

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

A backpack can hold at most 10 kilograms. Items: tent 6 kg, value 8; stove 4 kg, value 5; food 5 kg, value 6; lamp 1 kg, value 3. Which choice has the greatest total value?

Answer: A. Food, stove, lamp

Reasoning: Food, stove and lamp weigh exactly 10 kg and are worth 14. Taking the most valuable item (the tent) first leaves room for at most 4

more kg, and the best that adds is the stove: 13.

- ✗ Tent, stove: 10 kg with value 13, less than 14
- ✗ Tent, lamp: 7 kg with value 11
- ✗ Food, lamp: 6 kg with value 9

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S51-F1-Q5 · question page 1156

LOGIC.OPTIMIZE.01 · When the obvious choice is not best

Two whole numbers are each at least 1 and their product is 36. Make their sum as small as possible. Give the two numbers.

Answer: any values that satisfy: a == 6 and b == 6 (example: a = 6, b = 6)

Reasoning: The pairs are 1 x 36, 2 x 18, 3 x 12, 4 x 9, 6 x 6 with sums 37, 20, 15, 13, 12. 6 and 6 is best.

Tempting answers the rule rejects: a = 4, b = 9; a = 1, b = 36; a = 3, b = 12

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S51-F2-Q1 · question page 1157

LOGIC.OPTIMIZE.01 · Prove the best value

Whole numbers add up to 20 (you choose how many numbers). What is the largest possible product?

Answer: 1458

Reasoning: Any part 5 or more can be split into 2 and the rest to raise

the product, and 4 = 2 + 2 gives the same product, so use only 2s and 3s. Three 2s (product 8) are worse than two 3s (product 9), so use at most two 2s: 20 = 3 x 6 + 2, product 729 x 2 = 1458.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F2-Q2 · question page 1158

LOGIC.OPTIMIZE.01 · Prove the best value

Coins are worth 1, 7 and 10 cents. What is the fewest coins that make exactly 14 cents?

Answer: 2

Reasoning: Taking the largest coin first gives 10 + 1 + 1 + 1 + 1 (5 coins), but 7 + 7 uses 2, and no single coin is worth 14.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S51-F2-Q3 · question page 1159

LOGIC.OPTIMIZE.01 · Prove the best value

Among rectangles with whole-number sides and area 36, the 6 by 6 square has the smallest perimeter.

Answer: True

Reasoning: Perimeters: 1 x 36: 74, 2 x 18: 40, 3 x 12: 30, 4 x 9: 26, 6 x 6: 24.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S51-F2-Q4 · question page 1160

LOGIC.OPTIMIZE.01 · Prove the best value

Whole numbers a, b and c are each at least 1 and add up to 12. Make the product a x b x c as large as possible. Give a, b and c.

Answer: any values that satisfy: a == 4 and b == 4 and c == 4 (example: a = 4, b = 4, c = 4)

Reasoning: If two of the numbers differ by 2 or more, moving 1 from the larger to the smaller raises the product. So all three are equal: 4, 4, 4, product 64.

Tempting answers the rule rejects: a = 3, b = 4, c = 5; a = 2, b = 5, c = 5; a = 6, b = 6, c = 0

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all triples: the largest product is 64, reached only at (4, 4, 4). Automatic: the example passes the rule and 3 listed mistakes fail it.

S51-F2-Q5 · question page 1161

LOGIC.OPTIMIZE.01 · Prove the best value

Use each of the digits 1, 2, 3 and 4 once to make two 2-digit numbers. Which pair has the largest product?

Answer: D. 41 x 32

Reasoning: The tens digits should be 4 and 3. Then 41 x 32 = 1312 beats 42 x 31 = 1302: the extra unit goes with the smaller tens digit.

- ✗ 42 x 31: 42 x 31 = 1302, less than 1312
- ✗ 43 x 21: 43 x 21 = 903
- ✗ 31 x 24: 31 x 24 = 744

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-M-Q1 · question page 1163

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 3, 7, 11, 15, ...?

Answer: 19

Reasoning: Rule (1): add 4 each time. 15 + 4 = 19.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-M-Q2 · question page 1164

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 2, 6, 18, 54, ...?

Answer: A. Multiply by 3 each time

Reasoning: Each term is 3 times the one before.

- ✗ Add 4 each time: 2 + 4 = 6, but 6 + 4 = 10, not 18
- ✗ Add 4, then 12, then 36: those amounts do not grow by the same number each time
- ✗ Multiply by 2 each time: 2 x 2 = 4, not 6

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-M-Q3 · question page 1165

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 1, 2, 4, 7, 11, ...?

Answer: 16

Reasoning: Rule (3): the amounts added are 1, 2, 3, 4, so next add 5: 16.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-M-Q4 · question page 1166

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 5, 10, 20, 35, ... follows rule (3).

Answer: True

Reasoning: The amounts added are 5, 10, 15, which grow by 5 each time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-M-Q5 · question page 1167

LOGIC.PATTERN.01 · Identify the rule in a stated family that fits a sequence, and continue it

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters C, F, I, L follow rule (1) by their numbers. Give the next letter.

Answer: any values that satisfy: x == 'o' (example: x = O)

Reasoning: C, F, I, L are 3, 6, 9, 12: add 3. Next is 15, which is O.

Tempting answers the rule rejects: x = M; x = N; x = P

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: next number 15 is letter O. Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-B1-Q1 · question page 1168

LOGIC.PATTERN.01 · Strategy: write the differences

Write the differences of 2, 5, 9, 14. What is the last difference?

Answer: 5

Reasoning: The differences are 3, 4, 5.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-B1-Q2 · question page 1169

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 10, 7, 4, 1, ... follows rule (1).

Answer: True

Reasoning: It adds -3 each time.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-B1-Q3 · question page 1170

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Which rule fits 1, 4, 16, 64, ...?

Answer: B. Rule (2)

Reasoning: Each term is 4 times the one before.

- ✗ Rule (1): the differences 3, 12, 48 are not equal
- ✗ Rule (3): the differences 3, 12, 48 do not grow by the same amount
- ✗ None of the rules: multiplying by 4 each time works

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-B1-Q4 · question page 1171

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the next term of 50, 45, 40, 35, ...?

Answer: 30

Reasoning: Add -5: 30.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-B1-Q5 · question page 1172

LOGIC.PATTERN.01 · Strategy: write the differences

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence 3, 6, 12, 24, ... follows rule (1).

Answer: False

Reasoning: The differences are 3, 6, 12, which are not equal. It follows rule (2): multiply by 2.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-F1-Q1 · question page 1173

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. The letters B, D, G, K follow rule (3) by their numbers. Give the next letter.

Answer: any values that satisfy: x == 'p' (example: x = P)

Reasoning: B, D, G, K are 2, 4, 7, 11; the amounts added are 2, 3, 4, so add 5: 16, which is P.

Tempting answers the rule rejects: x = O; x = N; x = Q

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: next number 16 is letter P. Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-F1-Q2 · question page 1174

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow

by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 4, 7, 10, 13, ...?

Answer: 31

Reasoning: Rule (1), add 3. The 10th term is 4 + 9 x 3 = 31.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F1-Q3 · question page 1175

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Knowing only the first two terms, 3 and 6, is enough to know the third term.

Answer: False

Reasoning: Rule (1) gives 9 (add 3) and rule (2) gives 12 (multiply by 2), so more terms are needed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S52-F1-Q4 · question page 1176

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The first three terms are 1, 2, 4. Which statement is true?

Answer: C. Rule (2) and rule (3) both fit, giving 8 or 7 next.

Reasoning: Three terms cannot tell rule (2) from rule (3) here.

✗ Only rule (2) fits, so the next term is 8.: rule (3) also fits: add 1, then 2, then 3

✗ Only rule (3) fits, so the next term is 7.: rule (2) also fits: multiply by 2

✗ Rule (1) fits, so the next term is 6.: the differences 1 and 2 are not equal

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S52-F1-Q5 · question page 1177

LOGIC.PATTERN.01 · Rules for letters and far terms

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 8th term of 2, 6, 18, 54, ...?

Answer: 4374

Reasoning: Rule (2), multiply by 3. The 8th term is 2 x 3 to the 7th power = 4374.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q1 · question page 1178

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). The sequence is 5, 9, 13, 17, ... Give the 25th term.

Answer: any values that satisfy: t == 101 (example: t = 101)

Reasoning: Rule (1), add 4: the nth term is 5 + 4(n - 1). For n = 25: 5 + 96 = 101.

Tempting answers the rule rejects: t = 105; t = 97; t = 100

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Automatic: the example passes the rule and 3 listed mistakes fail it.

S52-F2-Q2 · question page 1179

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). In the sequence 3, 6, 12, 24, ..., which term is the first one greater than 1000? Give its position (1 for the first term).

Answer: 10

Reasoning: Terms: 3, 6, 12, 24, 48, 96, 192, 384, 768, 1536. The 10th term, 1536, is the first above 1000.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q3 · question page 1180

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). What is the 10th term of 1, 3, 6, 10, ...?

Answer: 55

Reasoning: Rule (3): add 2, 3, 4, ... The nth term is 1 + 2 + ... + n, so the 10th is 55.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S52-F2-Q4 · question page 1181

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). A sequence that follows rule (1) with a positive amount added will eventually pass every number.

Answer: True

Reasoning: Adding the same positive amount again and again grows without limit.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S52-F2-Q5 · question page 1182

LOGIC.PATTERN.01 · Work with the rule as a formula

Rule family: every sequence here follows exactly one of these rules. (1) Add the same number each time. (2) Multiply by the same number each time. (3) Add amounts that grow by the same number each time (for example +1, +2, +3, ...). Letters are numbered by their place in the alphabet: A = 1, B = 2, ..., Z = 26. Which letter sequence follows rule (2) by the letters' numbers?

Answer: D. A, B, D, H

Reasoning: A, B, D, H are 1, 2, 4, 8: multiply by 2.

- ✗ A, C, E, G: that adds 2 each time: rule (1)
- ✗ A, B, D, G: that adds 1, 2, 3: rule (3)
- ✗ B, C, E, H: that adds 1, 2, 3: rule (3)

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-M-Q1 · question page 1184

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: D. MELON

Reasoning: MELON uses M, E, L, O, N once each, like LEMON.

- ✗ SOLEMN: SOLEMN has an extra S
- ✗ LANE: LANE has different letters
- ✗ TONE: TONE has different letters

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-M-Q2 · question page 1185

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: any values that satisfy: w == 'tears' or w == 'rates' (example: w = TEARS)

Reasoning: TEARS and RATES use A, E, R, S, T.

Tempting answers the rule rejects: w = STARE; w = STAR; w = TASTE; w = ASTER

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked: anagrams of STARE on the list are ['TEARS', 'RATES']. Automatic: the example passes the rule and 4 listed mistakes fail it.

S53-M-Q3 · question page 1186

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: True

Reasoning: Each uses E, I, L, N, S, T once.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-M-Q4 · question page 1187

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be

Answer: 15

Reasoning: Check each word against P, L, A, N, E, T: PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL: 15.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-M-Q5 · question page 1188

LOGIC.WORD.03 · Find anagrams and letter-bank words within a supplied word list and letter rules

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Read the letters STOPAN from righ

Answer: B. POTS

Reasoning: Backward, STOPAN reads NAPOTS, which contains POTS.

- ✗ STOP: STOP appears forward, not in the backward reading NAPOTS
- ✗ SPOT: the letters S, P, O, T are not neighbors in that order in NAPOTS
- ✗ PAN: PAN is not on the list

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-B1-Q1 · question page 1189

LOGIC.WORD.03 · Strategy: sort the letters

Write the letters of NOTE in alphabetical order (no spaces).

Answer: ENOT

Reasoning: E, N, O, T.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-B1-Q2 · question page 1190

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. STONE and NOTES are anagrams.

Answer: True

Reasoning: Both sort to E, N, O, S, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-B1-Q3 · question page 1191

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. TASTE and STATE are anagrams.

Answer: True

Reasoning: Both sort to A, E, S, T, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-B1-Q4 · question page 1192

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. Which word is NOT an anagram of SPOT?

Answer: C. SPOTS

Reasoning: SPOTS has an extra S.

- ✗ STOP: STOP uses S, P, O, T
- ✗ TOPS: TOPS uses S, P, O, T
- ✗ POTS: POTS uses S, P, O, T

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-B1-Q5 · question page 1193

LOGIC.WORD.03 · Strategy: sort the letters

An anagram uses exactly the same letters, each as many times, in a different order. BREAD and BEARD are anagrams.

Answer: True

Reasoning: Both sort to A, B, D, E, R.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F1-Q1 · question page 1194

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Join one part from the first grou

Answer: any values that satisfy: w == 'planet' or w == 'heart' or w == 'stone' (example: w = PLANET)

Reasoning: PL + ANET = PLANET, HE + ART = HEART and ST + ONE = STONE are all on the list.

Tempting answers the rule rejects: w = START; w = PLONE; w = STANET

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked all 9 joins: listed words ['PLANET', 'STONE', 'HEART']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S53-F1-Q2 · question page 1195

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 2

Reasoning: HEART and HATER.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F1-Q3 · question page 1196

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. You

may use the letters S, T, O,

Answer: A. All of SPOT, STOP, POTS, TOPS and POST tie at 4 letters

Reasoning: Only 4-letter words from S, T, O, P are on the list, and there are five of them.

- ✗ STOP is the only longest word: SPOT, POTS, TOPS and POST also have 4 letters
- ✗ STONE: STONE needs N and E
- ✗ SPOTS: SPOTS is not on the list

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-F1-Q4 · question page 1197

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: True

Reasoning: Both sort to G, H, I, N, T.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F1-Q5 · question page 1198

LOGIC.WORD.03 · Letter banks and joined parts

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN,

LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 2

Reasoning: LISTEN, SILENT, ENLIST, TINSEL; SPOT, STOP, POTS, TOPS, POST; STONE, ONSET, NOTES (only 3). So 2 families.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F2-Q1 · question page 1199

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: C. TASTE PEN

Reasoning: PAST TEEN has A, E, E, N, P, S, T, T. TASTE (A, E, S, T, T) and PEN (E, N, P) use exactly these.

- ✗ PETAL NET: it has an L and no S
- ✗ STAR PEN: it has an R and only one T
- ✗ NEAT POTS: it has an O and only one E

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S53-F2-Q2 · question page 1200

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Every word on the list that is al

Answer: True

Reasoning: The pairs are NET and TEN, SPOT and TOPS, STOP and POTS, RATS and STAR.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S53-F2-Q3 · question page 1201

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. Give a word on the list that is a

Answer: any values that satisfy: w == 'net' or w == 'ten' or w == 'spot' or w == 'stop' or w == 'pots' or w == 'tops' or w == 'rats' or w == 'star' (example: w = STOP)

Reasoning: NET, TEN, SPOT, TOPS, STOP, POTS, RATS and STAR all work.

Tempting answers the rule rejects: w = POST; w = STONE; w = LEMON

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Checked every list word read backward: ['NET', 'TEN', 'SPOT', 'STOP', 'POTS',

'TOPS', 'RATS', 'STAR']. Automatic: the example passes the rule and 3 listed mistakes fail it.

S53-F2-Q4 · question page 1202

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. How many words on the list can be

Answer: 7

Reasoning: NOTE, TONE, STONE, ONSET, NOTES, NET, TEN: 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S53-F2-Q5 · question page 1203

LOGIC.WORD.03 · Phrases and hidden words

Word list (the ONLY words allowed): MELON, LEMON, SOLEMN, TEARS, RATES, STARE, TASTE, STATE, LISTEN, SILENT, ENLIST, TINSEL, PLANE, PANEL, PLANET, LEAN, LANE, NEAT, ANT, TAN, PEN, NET, TEN, PALE, LEAP, PLATE, PETAL, SPOT, STOP, POTS, TOPS, POST, RATS, STAR, ARTS, NIGHT, THING, DUSTY, STUDY, NOTE, TONE, STONE, ONSET, NOTES, BREAD, BEARD, BARED, HEART, EARTH, HATER. An anagram uses exactly the same

Answer: 3

Reasoning: STONE, ONSET and NOTES: 3.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-M-Q1 · question page 1205

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Answer: 2

Reasoning: Weigh 3 against 3. If one side goes down, the heavy coin is among those 3; if they balance, it is among the other 3. Then weigh 1 against 1 of the 3. One weighing cannot work: it has only 3 outcomes for 9 possibilities.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-M-Q2 · question page 1206

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 9 coins and one is heavier. Which first weighing lets you always finish in the fewest total weighings?

Answer: A. 3 coins against 3 coins

Reasoning: Splitting 9 into three groups of 3 leaves 3 suspects whatever happens, and 3 needs only one more weighing.

- ✗ 4 coins against 4 coins: if one side goes down, 4 suspects remain, which needs 2 more weighings
- ✗ 1 coin against 1 coin: if they balance, 7 suspects remain, which needs

2 more weighings

✗ 2 coins against 2 coins: if they balance, 5 suspects remain, which needs 2 more weighings

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason Automatic: the check function accepts only the keyed choice.

S54-M-Q3 · question page 1207

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 weighings you can always find the heavier coin among 27 coins.

Answer: True

Reasoning: Each weighing splits the suspects into three groups (left, right, off the scale): 27, then 9, then 3, then 1.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-M-Q4 · question page 1208

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. What is the largest number of coins n for which 2 weighings always find the one heavier coin?

Answer: any values that satisfy: n == 9 (example: n = 9)

Reasoning: Two weighings have 3 x 3 = 9 outcome patterns, so at most 9 coins; and 9 works by splitting into threes.

Tempting answers the rule rejects: n = 8; n = 10; n = 6

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: largest n with need(n) <= 2 is 9. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-M-Q5 · question page 1209

LOGIC.SEARCH.02 · Plan weighings or questions and find the least number that always identifies the answer

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 10 coins and one is heavier. What is the least number of weighings that always finds the heavy coin?

Answer: 3

Reasoning: Two weighings give only 9 outcome patterns, which is fewer than 10 possibilities, so 2 are not enough. Three are: weigh 3 against 3, leaving at most 4 suspects, then finish in 2 more.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q1 · question page 1210

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

How many different outcomes can one weighing on a balance scale have?

Answer: 3

Reasoning: Left side down, right side down, or balance.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q2 · question page 1211

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 3 coins and one heavier, one weighing always finds the heavy coin.

Answer: True

Reasoning: Weigh 1 against 1; a balance means the third coin is heavy.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-B1-Q3 · question page 1212

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 5 coins and one is heavier. You weigh 2 against 2 and they balance. What do you know?

Answer: B. The fifth coin is the heavy one.

Reasoning: All four weighed coins are genuine, so the coin left off is heavy.

- ✗ One of the four weighed coins is heavy.: a heavy coin on the scale would tip it
- ✗ Nothing yet.: a balance clears all four weighed coins
- ✗ You must weigh again to know.: only one coin is left as a suspect

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-B1-Q4 · question page 1213

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

How many outcome patterns do 2 weighings have in all?

Answer: 9

Reasoning: Each of the 3 outcomes of the first weighing can be followed by 3 outcomes of the second: 9.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-B1-Q5 · question page 1214

LOGIC.SEARCH.02 · Strategy: each weighing has three outcomes

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, one weighing always finds the heavy coin.

Answer: False

Reasoning: One weighing has 3 outcomes, but there are 4 possibilities.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-F1-Q1 · question page 1215

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. You may use 4 weighings. What is the largest number of coins among which you can always find the one heavier coin?

Answer: 81

Reasoning: 4 weighings have 3 x 3 x 3 x 3 = 81 outcome patterns, and

splitting into thirds each time handles 81 coins.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F1-Q2 · question page 1216

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 6 coins and one is heavier. You weigh 2 against 2 and the left side goes down. What should the next weighing be?

Answer: D. One left-side coin against the other left-side coin

Reasoning: Only the 2 left-side coins are suspects. Weigh them against each other: the one that goes down is heavy.

- ✗ One left-side coin against a right-side coin: the right-side coins are already known to be genuine, so this tests only one suspect
- ✗ The two coins left off against each other: those coins are genuine; the heavy coin was on the left
- ✗ All six coins again, 3 against 3: that repeats work and may not finish

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-F1-Q3 · question page 1217

LOGIC.SEARCH.02 · Bigger groups and comparisons

You have 3 weights of different sizes and compare them two at a time on a balance. Finding the heaviest always takes at least 2 weighings.

Answer: True

Reasoning: Each weight that is not the heaviest must lose at least one

comparison, and one weighing has only one loser. Two weights must lose, so 2 weighings are needed.

Grading: The chosen True/False must equal the key.

Verified: solved by the author

S54-F1-Q4 · question page 1218

LOGIC.SEARCH.02 · Bigger groups and comparisons

You have 8 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds the heaviest?

Answer: 7

Reasoning: Every weight except the heaviest must lose a comparison, and each weighing has one loser: at least 7. A knockout tournament uses exactly 7.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F1-Q5 · question page 1219

LOGIC.SEARCH.02 · Bigger groups and comparisons

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 30 for which 3 weighings always find the one heavier coin.

Answer: any values that satisfy: isInteger(n) and n >= 20 and n <= 27 (example: n = 27)

Reasoning: Three weighings handle up to 3 x 3 x 3 = 27 coins, so any n from 20 to 27 works; 28, 29 and 30 do not.

Tempting answers the rule rejects: n = 28; n = 30; n = 19

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: need(n) <= 3 exactly for n up to 27. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-F2-Q1 · question page 1220

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. There are 7 coins and one is LIGHTER (not heavier). What is the least number of weighings that always finds it?

Answer: 2

Reasoning: The same splitting works for a lighter coin: weigh 3 against 3 (or 2 against 2), leaving at most 3 suspects; then one more weighing. One weighing cannot tell 7 cases apart.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.

S54-F2-Q2 · question page 1221

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Why can 2 weighings NOT always find the heavy coin among 10?

Answer: A. Two weighings give only 9 outcome patterns, but there are 10 possible heavy coins.

Reasoning: Each possibility needs its own outcome pattern, and 10 > 9.

✗ 10 is an even number.: evenness is not the issue; 8 coins need only 2 weighings

✗ You cannot put 5 coins on each side.: you can, and it still would not be enough

✗ The first weighing must use all 10 coins.: it need not use every coin

Grading: The chosen letter must equal the keyed letter.

Verified: solved by the author; each wrong choice checked against its reason

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S54-F2-Q3 · question page 1222

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. With 4 coins and one heavier, 2 weighings always find the heavy coin.

Answer: True

Reasoning: Weigh 1 against 1. If one goes down, done; if they balance, weigh the other two.

Grading: The chosen True/False must equal the key.

Verified: solved by the author Automatic: the statement was evaluated by code.

S54-F2-Q4 · question page 1223

LOGIC.SEARCH.02 · Least weighings, proved

A balance scale compares two groups of coins: one side goes down, or the two sides balance. All coins look the same. All genuine coins weigh the same; the one odd coin is heavier unless the question says otherwise. Give a number of coins n from 20 to 90 for which 3 weighings are NOT enough but 4 weighings are enough to always find the one heavier coin.

Answer: any values that satisfy: isInteger(n) and n >= 28 and n <= 81 (example: n = 50)

Reasoning: 3 weighings handle at most 27 coins, and 4 handle at most 81. So n from 28 to 81 works.

Tempting answers the rule rejects: n = 27; n = 82; n = 20

Grading: The learner's values are put into the rule; ANY values that make the rule true are correct.

Verified: Search: need(n) == 4 exactly for n from 28 to 81. Automatic: the example passes the rule and 3 listed mistakes fail it.

S54-F2-Q5 · question page 1224

LOGIC.SEARCH.02 · Least weighings, proved

You have 4 weights of different sizes and compare them two at a time on a balance. What is the least number of weighings that always finds both the heaviest and the lightest?

Answer: 4

Reasoning: Compare A with B and C with D. The heaviest is the winner of a second comparison between the two winners, and the lightest is the loser of a comparison between the two losers: 4 weighings. A search of every strategy shows 3 are never enough.

Grading: The game's answer checker compares the typed value with the key (equivalent number forms accepted).

Verified: solved by the author Automatic: the key was recomputed by code.